Testing Mediation Effects: A Theoretical Case for Bootstrap Percentile Confidence Interval

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Abstract

Traditionally, the theoretical justification for bootstrap confidence intervals (C.I.) relies on the Gaussian asymptotic approximation of the underlying point estimator. Nevertheless, bootstrap C.I.s are frequently used in empirical work even when the point estimator is not asymptotically normal. Specifically, this is the case for testing the significance of a mediation channel in causal analysis. This paper shows that the usage of bootstrap percentile C.I.s (Hansen (2022)) for testing the mediation effect can be justified even when an asymptotically Gaussian estimator is not available. To do so, we use a novel theoretical framework in which we show that the coverage probability converges to an event probability of the limit conditional cumulative distribution function (CDF) of the bootstrap statistic. The CDF has a closed form solution. Using the solution, we show that the percentile C.I. is asymptotically valid in an empirically relevant case and also propose a simple modification that yields more power in that case, as well as asymptotic validity more generally.

1 Introduction

Since the seminal work of Efron (1979), bootstrap methods have become a standard tool for statistical inference, particularly when analytic approximations are complicated or unavailable. In particular, the bootstrap percentile interval (PCI) and tests based on PCI are especially popular due to their simplicity and ease of implementation (Hansen, 2022).

However, traditionally, the theory for bootstrap inference has largely been developed for regular settings in which the distribution of the original sample test statistic is consistently estimated by that of its bootstrap

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version (see Horowitz (2001) and references therein). Within this framework, the PCI has often been viewed less favorably than alternatives such as the basic bootstrap or the bootstrap-t methods. For example, Hall (1988) argues that the PCI effectively uses bootstrap quantiles in the wrong direction and may perform poorly when the sampling distribution is asymmetric.

Whether such intuition and recommendations extend to nonregular problems is not clear. Indeed, numerical evidence suggests that PCI often performs remarkably well even in settings with substantial skewness. For example, Hesterberg (2015) and Efron and Tibshirani (1994) (Section 13.4-5) report examples in which PCI outperforms the basic bootstrap and the bootstrap-t confidence intervals.

Since the standard bootstrap asymptotic theory fails to provide justification in nonregular problems, we adopt an alternative asymptotic framework. In this framework, we establish the weak convergence of the random conditional law of the bootstrap statistic to a random law. Using this, we express the limit of the coverage probability of the PCI as an event probability regarding the limit conditional CDF of the bootstrap statistic. Studying the distribution of this conditional CDF allows us to gain a closer look at the coverage probability than the standard bootstrap asymptotic theory would.

We apply it to testing a hypothesis of the form: $H_0 : \theta_1\theta_2 = 0$. In this setting, even though θ_1 and θ_2 typically have asymptotically jointly normal estimators, the parameter of interest $\theta_1\theta_2$ may not. This occurs because H_0 may hold both when exactly one coefficient is zero and when both coefficients are zero. The two cases lead to different asymptotic behaviors of the plug-in estimator of $\theta_1\theta_2$. When one coefficient is zero and the other is bounded away from zero, the product estimator admits a conventional normal approximation through the delta method. In contrast, when both coefficients are zero or drift to zero faster than a certain rate, the first-order derivative of the product map vanishes, and the limiting distribution is no longer normal. Conventional bootstrap asymptotic theory is not enough to elucidate the behavior of bootstrap procedures in the latter case because the conditional limit distribution of the bootstrap statistic differs from the limit distribution of the original sample statistic.

A leading application of the hypothesis $H_0 : \theta_1\theta_2 = 0$ is testing the significance of a mediation channel in causal inference (ref. Baron and Kenny (1986)). In the standard linear mediation model, the effect of the treatment variable on a mediator (M) is denoted θ_1, and the effect of the mediator on the outcome is denoted θ_2. Thus, no mediation effect through M is expressed as $\theta_1\theta_2 = 0$. Mediation analysis has been widely applied across disciplines, including behavioral science, communication, and medicine, as well as in economics. For a comprehensive review of the applications in economics see Huber (2020).

Several methods have been proposed for testing this hypothesis, including the Delta method-based t-test called the Sobel test (Sobel (1982)) and various bootstrap methods. Bootstrap methods are often recommended in practice because simulation studies find them delivering higher power than the Sobel test

in the nonregular case where both θ_1 and θ_2 are small. See e.g. MacKinnon et al. (2004); Fritz and MacKinnon (2007); Hayes (2009); Barfield et al. (2017). Some specifically recommend the PCI test, which rejects H_0 when the PCI excludes 0 (Hayes and Rockwood (2017)). However, the theoretical foundation for these recommendations is lacking. Existing theoretical analyses do not support these recommendations (He et al. (2023)).[1]

Our analysis provides the first theoretical justification for the popularity of the PCI test under one condition. The condition is that the estimators of θ_1 and θ_2 are asymptotically uncorrelated. We prove that the PCI test is asymptotically valid under drifting sequences of parameters where θ_1 or θ_2 can drift to zero at any rate or not at all. Hence, it rejects with asymptotic probability at most equal to the nominal level under $H_0 : \theta_1\theta_2 = 0$, regardless of whether both parameters are zero, one is zero and the other is close to zero, or one is zero and the other is far from zero.

The analysis also offers the first theoretical explanation for a well documented over-coverage of the PCI when $\theta_1 = \theta_2 = 0$, for example, by MacKinnon et al. (2004). We derive an analytical formula for the asymptotic probability that the PCI covers 0 in this case. The formula helps us construct a refinement that achieves higher power while maintaining size control. The refined PCI is always a subset of the original PCI. It is constructed by adjusting the quantile level in a data-driven manner. The adjustment has a closed-form formula and is easy to compute.

The asymptotically uncorrelated case is an important one in practice. For example, it arises when the linear structural equation model is correctly specified. It is also implied by the sequential ignorability condition introduced by Imai et al. (2010). Imai et al. (2010) show that this condition is sufficient for $\theta_1\theta_2$ to be interpreted as the average mediation effect; See also van Garderen and van Giersbergen (2024b).

Nonetheless, $H_0 : \theta_1\theta_2 = 0$ is sometimes tested when the estimators for θ_1 and θ_2 are not asymptotically independent. We show that the PCI and its refined version under-cover 0 when H_0 is true if the estimators exhibit high enough correlation. We find the under-coverage to be noticeable when the absolute value of the correlation coefficient is greater than 0.9 for the PCIs with nominal coverage 90%. To remedy this problem, we propose a correlation-dependent size correction factor that widens the refined PCI. We provide a table of the correction factor for a grid of correlation values. Correcting the refined PCI test using this table restores its asymptotic validity even when the correlation is large.

Our Monte Carlo simulations support the theoretical findings. The PCI and our refined version both control size well at moderate correlation values, while the refined procedure achieves higher power. The size-corrected refined PCI controls size regardless of the correlation value. In contrast, several competing

[1] They use standard bootstrap asymptotic theory and show that the limit conditional distribution of the bootstrap statistic is not the same as the limit distribution of the original sample statistic. They use the difference as evidence against the basic bootstrap; the same argument would apply to PCI as well.

bootstrap methods, including the basic bootstrap, the bias corrected (BC) and bias corrected and accelerated (BCa) bootstrap test, the bootstrap-t test, and the adaptive bootstrap test (AB) in He et al. (2023) exhibit size distortions when one coefficient is zero and the other is close to zero. The nearly similar test proposed in van Garderen and van Giersbergen (2024b) has size distortion when the correlation of the estimators of θ_1 and θ_2 is nonzero.

We apply the PCI and the refined PCI to the data set from the Jobs II Preventive Intervention for Unemployed Job Seekers, 1991-1993. We compute confidence intervals for the mediation effect of job-search self-efficacy in job training on employment outcomes. We find that our refined PCI can be meaningfully narrower than the raw PCI and the asymptotic confidence interval.

We are not the first to obtain positive results regarding bootstrap tests when the conditional distribution of bootstrap statistics converges weakly to a random limit. Cavaliere and Georgiev (2020) show that the basic bootstrap p-values are valid when there is a conditioning object (measurable with respect to the original sample) given which the distribution of the original sample test statistic converges to the same random limit as the conditional distribution of the bootstrap statistic. For example, the sample of the non-stationary regressors serves as this conditioning object in a linear regression with non-stationary regressors. The mediation test problem does not fit into this framework because such a conditioning object is not available.

This paper is also related to the recent paper by Miller et al. (2026). Miller et al. focus on a closely related null hypothesis: $H_0 : \theta_1\theta_2 \geq 0$. They also prove a validity result for a PCI based test, which they call the Kowalski (2023) heuristic bootstrap test, in the case where the correlation between the estimators of θ_1 and θ_2 is zero. However, their result is derived assuming a perfect bootstrap approximation for the limit distribution of the θ estimators and thus avoids the asymptotic derivations for bootstrap. They also do not derive a closed-form solution for the limit rejection probability and do not refine or size-correct the bootstrap test.

The remainder of the paper is organized as follows. Section 2 explains the PCI and the mediation model. Section 3 presents the asymptotic framework for studying the percentile method and analyzes its size properties. Section 4 introduces our refined percentile method and examines its size properties. Section 5 reports the Monte Carlo simulation results. Section 6 reports the empirical application. Section 7 concludes.

2 Setup

We consider testing the null hypothesis:

$$H_0 : \vartheta \equiv \theta_1\theta_2 = 0, \tag{1}$$

where θ_1 and θ_2 are two parameters that admit asymptotically jointly normally distributed estimators, denoted $\widehat{\theta}_{1n}$ and $\widehat{\theta}_{2n}$, where n indicates the sample size. This hypothesis is a composite hypothesis in that it holds as long as one of θ_1 and θ_2 is zero.

In the linear structural equation model (LSEM) for mediation analysis, θ_1 and θ_2 are, respectively, the coefficient on the treatment variable (X) in the mediator equation and the coefficient on the mediator (M) variable in the outcome (Y) equation:

$$M = \alpha_M + \theta_1 X + \delta_M' W + u, \qquad E[u(1, X, W')'] = \mathbf{0}, \tag{2}$$

$$Y = \alpha_Y + \theta_2 M + \beta X + \delta_Y' W + e, \qquad E[e(1, M, X, W')'] = \mathbf{0}, \tag{3}$$

where α_M and α_Y are intercepts, β is the direct effect of X on Y, δ_M and δ_Y are coefficients on the control variables W, and u and e are the structural error terms. The product $\vartheta \equiv \theta_1\theta_2$ is taken as the mediation effect of X on Y through M. Rejecting the null hypothesis in (1) indicates that the mediation channel M is statistically significant. The LSEM is widely used in practice, and $H_0 : \theta_1\theta_2 = 0$ is frequently tested. The latter is reflected in the more than 20,000 Google Scholar citations of Sobel (1982). Typically, $\widehat{\theta}_1$ and $\widehat{\theta}_2$ are ordinary least squares estimators (OLS) based on the regression equations above. [2]

2.1 Percentile Confidence Interval

Let the plug-in estimator of $\theta_1\theta_2$ be denoted as $\widehat{T}_n$; that is,

$$\widehat{T}_n = \widehat{\theta}_{1n}\widehat{\theta}_{2n}. \tag{4}$$

Let the bootstrap version of $\widehat{T}_n$ be denoted as $\widehat{T}_n^*$:

$$\widehat{T}_n^* = \widehat{\theta}_{1n}^*\widehat{\theta}_{2n}^* \tag{5}$$

where $\widehat{\theta}_{1n}^*$ and $\widehat{\theta}_{2n}^*$ are calculated in the same way as $\widehat{\theta}_{1n}$ and $\widehat{\theta}_{2n}$ except using a bootstrap sample.[3]

Let the sample we use to calculate $\widehat{\theta}_{1n}$ and $\widehat{\theta}_{2n}$ be called the *original sample*, and let $\mathcal{F}_n$ denote the σ-field generated by the original sample. For any $p \in (0, 1)$, define the $100p\%$ conditional bootstrap quantile by

$$\widehat{Q}_{\widehat{T}_n^*}(p) \quad = \inf\{t \in \mathbb{R} : \Pr(\widehat{T}_n^* \leq t | \mathcal{F}_n) \geq p\}.$$

[2] Imai et al. (2010) establishes conditions under which the above interpretation of ϑ is valid. We do not discuss the causal interpretation in this paper but only focus on the statistical aspect of testing $H_0 : \vartheta = 0$.

[3] The bootstrap sampling mechanism can be that of the non-parametric bootstrap, parametric bootstrap, residual bootstrap, multiplier bootstrap, block bootstrap, etc, as long as it ensures that the assumptions we impose on $\widehat{\theta}_{1n}^*$ and $\widehat{\theta}_{2n}^*$ are satisfied.

In practice, these quantiles are approximated by the empirical quantiles of a large number of independent copies of $\widehat{T}_n^*$ from many bootstrap replications. The $100(1-\alpha)\%$ PCI of ϑ is then defined as

$$PCI_{n,1-\alpha} = \left[\widehat{Q}_{\widehat{T}_n^*}\left(\frac{\alpha}{2}\right), \widehat{Q}_{\widehat{T}_n^*}\left(1-\frac{\alpha}{2}\right)\right]. \tag{6}$$

The PCI based test for the null hypothesis in (1) is to reject if $PCI_{n,1-\alpha}$ does not include 0:

$$\varphi_n^{PCI}(\alpha) = 1\{0 \notin PCI_{n,1-\alpha}\}. \tag{7}$$

The probability that $PCI_{n,1-\alpha}$ covers 0 is one minus the probability that $\varphi_n^{PCI}(\alpha)$ rejects H_0. Thus, in the subsequent text, when we discuss coverage probabilities, the implication on the rejection probabilities should be understood, and vice versa.

We also note that we only consider the use of the PCI to test H_0 in this paper and, hence, are only concerned about its coverage of 0.

3 Asymptotic Theory

3.1 The Degeneracy

We make the standard assumptions about the estimators $\widehat{\theta}_{1n}$ and $\widehat{\theta}_{2n}$ and their bootstrap version, which are easily satisfied by OLS estimators as well as other extremum estimators:

Assumption 1. (a) $\sqrt{n}\begin{pmatrix}\widehat{\theta}_{1n}-\theta_1\\ \widehat{\theta}_{2n}-\theta_2\end{pmatrix} \xrightarrow{d} \begin{pmatrix}\sigma_1 Z_1\\ \sigma_2 Z_2\end{pmatrix} \sim N(\mathbf{0}, \Sigma)$ *as* $n \to \infty$, *where* $\Sigma = \begin{pmatrix}\sigma_1^2 & \rho\sigma_1\sigma_2\\ \rho\sigma_1\sigma_2 & \sigma_2^2\end{pmatrix}$ *with* $\sigma_1 > 0, \sigma_2 > 0$ *and* $\rho \in [-1, 1]$.

(b) $\sqrt{n}\begin{pmatrix}\widehat{\theta}_{1n}^*-\widehat{\theta}_{1n}\\ \widehat{\theta}_{2n}^*-\widehat{\theta}_{2n}\end{pmatrix} \to_{d^*} \begin{pmatrix}\sigma_1 Z_1^*\\ \sigma_2 Z_2^*\end{pmatrix} \sim N(0, \Sigma)$, *where* $X_n^* \to_{d^*} X^*$ *means that* $\sup_{f \in BL_1(\mathbb{R}^2)} |\mathbb{E}[f(X_n^*)|\mathcal{F}_n] - \mathbb{E}[f(X^*)]| \to_p 0$, *with* $BL_1(\mathbb{R}^d) := \{f : \mathbb{R}^d \to \mathbb{R} : \sup_{x \in \mathbb{R}^d} |f(x)| \leq 1, |f(x_1) - f(x_2)| \leq \|x_1 - x_2\|$ *for all* $x_1, x_2 \in \mathbb{R}^d\}$ *being the bounded Lipschitz function space on* $\mathbb{R}^d$ *for a positive integer* d.

While the limit distribution of $(\widehat{\theta}_{1n}, \widehat{\theta}_{2n})$ is standard, that of the product $\widehat{\theta}_{1n}\widehat{\theta}_{2n}$ has a potential degeneracy under H_0. This is because the Jacobian matrix of the function $\theta_1\theta_2$ with respect to $(\theta_1, \theta_2)'$ is $\partial(\theta_1\theta_2)/\partial((\theta_1, \theta_2)') = (\theta_2, \theta_1)'$, and it is the zero vector when $\theta_1 = \theta_2 = 0$. By Delta method,

$$\sqrt{n}(\widehat{\theta}_{1n}\widehat{\theta}_{2n} - \theta_1\theta_2) \to_d N(0, \theta_2^2\sigma_1^2 + 2\rho\sigma_1\sigma_2\theta_1\theta_2 + \theta_1^2\sigma_2^2). \tag{8}$$

This limit distribution is degenerate when $\theta_1 = \theta_2 = 0$: it is a point mass at 0. Standard asymptotics theory for bootstrap C.I.s (ref. Hall (1992) and Horowitz (2001)) aims to prove that the conditional quantiles of

$\sqrt{n}(\widehat{\theta}^*_{1n}\widehat{\theta}^*_{2n} - \widehat{\theta}_{1n}\widehat{\theta}_{2n})$ converge to the corresponding quantiles of the limit distribution of $\sqrt{n}(\widehat{\theta}_{1n}\widehat{\theta}_{2n} - \theta_1\theta_2)$, and to argue that such convergence ensures correct asymptotic coverage probability. The argument relies on the continuity of the limit distribution at the appropriate quantiles. It fails when the limit distribution is degenerate.[4]

The problem exists not only in the knife edge case $\theta_1 = \theta_2 = 0$, but also when $\theta_1 = 0$ (or $\theta_2 = 0$) and θ_2 (or θ_1) is close to 0. In this case, the distribution of $\sqrt{n}(\widehat{\theta}_{1n}\widehat{\theta}_{2n} - \theta_1\theta_2)$ is nearly degenerate, the standard bootstrap theory provides a poor approximation of the actual finite sample behavior of the bootstrap C.I.s. Thus, it cannot be relied upon for the justification of these C.I.s.

3.2 Alternative Asymptotic Theory: Heuristic Derivation

In order to make progress, we first observe that the coverage probability of the PCI is (approximately) an event probability about the conditional CDF of $n\widehat{T}^*_n/(\sigma_1\sigma_2)$ at 0. Specifically,

$$\begin{aligned} \Pr(0 \in PCI_{n,1-\alpha}) &= \Pr\left(\widehat{Q}_{\widehat{T}^*_n}\left(\frac{\alpha}{2}\right) \leq 0 \leq \widehat{Q}_{\widehat{T}^*_n}\left(1-\frac{\alpha}{2}\right)\right) \\ &\approx \Pr\left(\frac{\alpha}{2} \leq \widehat{F}^*_n(0) \leq 1-\frac{\alpha}{2}\right), \end{aligned} \tag{9}$$

where $\widehat{F}^*_n(t) = \Pr(n\widehat{T}^*_n/(\sigma_1\sigma_2) \leq t|\mathcal{F}_n)$ and the $\approx$ is made precise in our formal proofs. This probability can be evaluated by studying the distribution of the random variable $\widehat{F}^*_n(0)$ directly. To do so, note that

$$n\widehat{T}^*_n = n\widehat{\theta}^*_{1n}\widehat{\theta}^*_{2n} = (\sqrt{n}(\widehat{\theta}^*_{1n} - \widehat{\theta}_{1n}) + \sqrt{n}(\widehat{\theta}_{1n} - \theta_1) + \sqrt{n}\theta_1) \cdot (\sqrt{n}(\widehat{\theta}^*_{2n} - \widehat{\theta}_{2n}) + \sqrt{n}(\widehat{\theta}_{2n} - \theta_2) + \sqrt{n}\theta_2). \tag{10}$$

Let $Z^*_{jn} = \sqrt{n}(\widehat{\theta}^*_{jn} - \widehat{\theta}_{jn})/\sigma_j$, $Z_{jn} = \sqrt{n}(\widehat{\theta}_{jn} - \theta_j)/\sigma_j$, and $h_{jn} = \sqrt{n}\theta_j/\sigma_j$. We can write $n\widehat{T}^*_n$ as

$$n\widehat{T}^*_n = \sigma_1\sigma_2(Z^*_{1n} + Z_{1n} + h_{1n})(Z^*_{2n} + Z_{2n} + h_{2n}). \tag{11}$$

Then, under Assumption 1, we show that the conditional law of $n\widehat{T}^*_n/(\sigma_1\sigma_2)$ given $\mathcal{F}_n$ as a random measure converges in law to the conditional law of the following random variable

$$T_h = (Z^*_1 + Z_1 + h_1)(Z^*_2 + Z_2 + h_2), \tag{12}$$

given (Z_1, Z_2) if $h_{1n} \to h_1$ and $h_{2n} \to h_2$, where (Z^*_1, Z^*_2) and (Z_1, Z_2) have the distributions specified in Assumption 1 and are independent of each other.

[4] Using a larger scaling constant, in this case, n, instead of $\sqrt{n}$ can make the limit distribution of $n(\widehat{\theta}_{1n}\widehat{\theta}_{2n} - \theta_1\theta_2)$ nondegenerate, but the limit distribution then depends on $\sqrt{n}\theta_j$ for $j = 1, 2$. The conditional distribution of $n(\widehat{\theta}^*_{1n}\widehat{\theta}^*_{2n} - \widehat{\theta}_{1n}\widehat{\theta}_{2n})$ depends on $\sqrt{n}\widehat{\theta}_j$, $j = 1, 2$ which differs from $\sqrt{n}\theta_j$, $j = 1, 2$ by a non-vanishing amount.

Therefore,

$$\Pr(0 \in PCI_{n,1-\alpha}) \approx \Pr\left(\frac{\alpha}{2} \leq F_h(0|Z) \leq 1 - \frac{\alpha}{2}\right). \tag{13}$$

where

$$F_h(s \mid Z) := \Pr(T_h \leq s \mid Z) = \Pr((Z_1^* + Z_1 + h_1)(Z_2^* + Z_2 + h_2) \leq s \mid Z), \qquad s \in \mathbb{R}. \tag{14}$$

The random variable $F_h(0|Z)$ depends on ρ, h_1, and h_2. Given a value of (ρ, h_1, h_2), one can easily derive its mathematical expression using properties of bivariate normal distributions.

$$\begin{aligned} F_h(0|Z) &= \Pr(Z_1^* + Z_1 + h_1 \geq 0, Z_2^* + Z_2 + h_2 \leq 0 | Z_1, Z_2) \\ &\quad + \Pr(Z_1^* + Z_1 + h_1 \leq 0, Z_2^* + Z_2 + h_2 \geq 0 | Z_1, Z_2) \\ &= \Phi(Z_1 + h_1) + \Phi(Z_2 + h_2) - 2\Phi_2(Z_1 + h_1, Z_2 + h_2, \rho), \end{aligned} \tag{15}$$

where $\Phi(\cdot)$ is the marginal CDF of the standard normal random variables $-Z_1^*$ and $-Z_2^*$, and $\Phi_2(\cdot, \cdot, \rho)$ is the joint CDF of $-Z_1^*$ and $-Z_2^*$ which have a correlation coefficient ρ. The closed-form expression for $F_h(0|Z)$ is then used to study the interval probability $\Pr(\alpha/2 \leq F_h(0|Z) \leq 1 - \alpha/2)$.

We find that the coverage probability is bounded below by $1 - \alpha$ when $\rho = 0$, and it can fall below $1 - \alpha$ for ρ's with large absolute values. We present our formal results in the next two subsections for the two cases.

3.3 Asymptotic Validity in the $\rho = 0$ Case

We start with $\rho = 0$. This is an empirically relevant case in mediation analysis. It holds for the model defined in (2)-(3) if $E[e|W, M, X] = 0$ when $\widehat{\theta}_1$ and $\widehat{\theta}_2$ are the OLS estimators.[5] The mean-independence is implied by the sequential ignorability assumption in Imai et al. (2010) (Theorem 2) under which they show that $\theta_1\theta_2$ identifies the average causal mediation effect.

Our formal asymptotic results allow θ_1 and θ_2 to drift with n as we let n go to infinity. This strictly generalizes asymptotic results derived while holding θ_1 and θ_2 fixed. By allowing the nonzero member of $\{\theta_1, \theta_2\}$ to drift to zero as $n \to \infty$, we obtain a limiting distribution that provides a better approximation to the finite sample distribution of the test statistic when the nonzero member of $\{\theta_1, \theta_2\}$ is small in absolute

[5] In the model without control variables and with the extra assumption, $E[u|X] = 0$, the asymptotic independence of $\widehat{\theta}_1$ and $\widehat{\theta}_2$ is proved in van Garderen and van Giersbergen (2024b) (Proposition 1). In Lemma 2 in the appendix, we show that the asymptotic independence holds even when there are control variables and $E[u|X] = 0$ is not imposed.

value.

Thus, from now on, we make the dependence of the population parameters θ_1 and θ_2 on n explicit by writing them as θ_{1n} and θ_{2n}.

The following result characterizes the asymptotic coverage probability of 0 for the PCI. Here we use the convention $\Phi(-\infty) = 0$ and $\Phi(\infty) = 1$.

Theorem 1. *Consider a sequence of data generating processes along which* $\sqrt{n}\theta_{1n}/\sigma_1 \to h_1 \in [-\infty, \infty]$, $\sqrt{n}\theta_{2n}/\sigma_2 \to h_2 \in [-\infty, \infty]$, *at least one of* h_1, h_2 *is zero, and Assumption 1 holds with* $\rho = 0$. *Let* $\alpha \in (0, 1)$. *Suppose, without loss of generality, that* $h_1 = 0$. *Then*

$$\lim_{n\to\infty} \Pr(0 \in PCI_{n,1-\alpha}) = E[p(\Phi(Z_2 + h_2))], \tag{16}$$

where $Z_2 \sim N(0, 1)$ *and* $p(v) = \begin{cases} \frac{1-\alpha}{1-2v} & v \in [0, \alpha/2) \\ 1 & v \in [\alpha/2, 1-\alpha/2] \\ \frac{1-\alpha}{2v-1} & v \in (1-\alpha/2, 1] \end{cases}$. *If, instead,* $h_2 = 0$, *then* $\lim_{n\to\infty} \Pr(0 \in PCI_{n,1-\alpha}) = E[p(\Phi(Z_1 + h_1))]$, *where* $Z_1 \sim N(0, 1)$.

Theorem 1 has three implications:

1. The PCI asymptotically over-covers in the degenerate case $\theta_1 = \theta_2 = 0$. In this case, $h_1 = h_2 = 0$, and it follows from the theorem that

$$\begin{aligned} \lim_{n\to\infty} \Pr(0 \in PCI_{n,1-\alpha}) = E[p(\Phi(Z_2))] &= \int_0^{\alpha/2} \frac{1-\alpha}{1-2v} dv + (1-\alpha) + \int_{1-\alpha/2}^1 \frac{1-\alpha}{2v-1} dv \\ &= (1-\alpha)\log(1-2v)|_{\alpha/2}^0 + 1 - \alpha = (1-\alpha)(1-\log(1-\alpha)), \end{aligned} \tag{17}$$

   where log stands for the natural logarithm function. Since $\log(1-\alpha) < 0$, we have $(1-\alpha)(1-\log(1-\alpha)) > 1-\alpha$. It in fact can be substantially larger than $1-\alpha$. For example, it is 0.9948, 0.9987, and 0.9999 when $1-\alpha = 0.9, 0.95$, and 0.99, respectively. This is consistent with observations in the literature based on Monte Carlo evidence; see e.g., Barfield et al. (2017). While the over-coverage of the PCI in the degenerate case is not new to practitioners, the derivation above is the first analytical proof.

2. The PCI has exact asymptotic coverage in the nondegenerate case. That is, when $\theta_1 = 0$ (or $\theta_2 = 0$) and θ_2 (or θ_1) is fixed at a nonzero value. This is because, in this case, $|h_2| = \infty$, and $\Phi(Z_2 + h_2) = 0$ or 1 almost surely. Thus, $p(\Phi(Z_2 + h_2)) = 1 - \alpha$ almost surely. This case is covered by standard bootstrap asymptotic theory which yields the same asymptotic result.

3. The PCI test is asymptotically valid regardless of whether we are in a degenerate case, a nondegenerate case, or cases in between. This is true because $p(v) \geq 1-\alpha$ for all $v \in [0,1]$, and $v \in [0,1]$ accommodates all h_2 values in $[-\infty, \infty]$. The value $h_2 = 0$ represents a small θ_{2n}, which is the degenerate case; $h_2 = \infty$ or $-\infty$ represents relatively large θ_{2n}, which is the non-degenerate case , and $h_2 \in \mathbb{R}$ represents cases in between, which we call nearly degenerate cases.

3.4 Potential Invalidity When ρ is not 0

While $\rho = 0$ is a common case in practice, sometimes the hypothesis H_0 is tested when $\rho \neq 0$. We show in this section that the PCI necessarily under-covers in cases that are degenerate or sufficiently nearly degenerate (i.e. $\max\{|h_1|, |h_2|\}$ is small enough) when $|\rho|$ is large enough.

Theorem 2. *Consider a sequence of data generating processes along which* $\sqrt{n}\theta_{1n}/\sigma_1 \to h_1 \in \mathbb{R}$, $\sqrt{n}\theta_{2n}/\sigma_2 \to h_2 \in \mathbb{R}$, $h_1 h_2 = 0$, *and Assumption 1 holds. Consider* $\alpha \in (0,1)$. *Without loss of generality, suppose that* $h_1 = 0$. *Then*

(a) $\lim_{n\to\infty} \Pr(0 \in PCI_{n,1-\alpha}) = \Pr(\Phi(Z_1) + \Phi(Z_2 + h_2) - 2\Phi_2(Z_1, Z_2 + h_2, \rho) \in (\alpha/2, 1-\alpha/2))$.

(b) $\lim_{\rho\to\pm 1} \lim_{n\to\infty} \Pr(0 \in PCI_{n,1-\alpha}) = \Pr\left(|\Phi(Z_1) - \Phi(Z_1 \pm h_2)| \in (\alpha/2, 1-\alpha/2)\right)$. *In particular, if* $|h_2| \leq 2\Phi^{-1}\left(\frac{1}{2} + \frac{\alpha}{4}\right)$, *the limit coverage probability is zero.*

To our knowledge, Theorem 2(b) is the first theoretical result that characterizes cases where the PCI under-covers.

Theorem 2(b) does not answer the question of how large $|\rho|$ has to be for the PCI to under-cover by a meaningful amount. To answer this question, we numerically calculate the limit coverage probability using the formula in part (a) by simulating (Z_1, Z_2) from $N\left(\mathbf{0}, \left(\begin{smallmatrix} 1 & \rho \\ \rho & 1 \end{smallmatrix}\right)\right)$, for a dense grid of (ρ, h_2) values on $(-1,1) \times \mathbb{R}$. Table 1 reports the numerically calculated range of ρ's for which the minimum limit coverage probability over the h_2 grid exceeds the nominal level up to a tolerance of 0.005. The results suggest that the 90% C.I. is asymptotically valid up to the stated tolerance for $|\rho| \leq 0.9$, the 95% C.I. for $|\rho| \leq 0.93$, and the 99% C.I. for $|\rho| \leq 0.97$. This result is encouraging for users of PCIs since it may be rare to have correlations between the estimators of θ_1 and θ_2 outside of these ranges in practice.

4 Refined PCI

Overall, the previous section reports good news about the PCI test for H_0: it is asymptotically valid regardless of degeneracy for a broad range of ρ values. However, it also reveals some areas for improvement: the test

Table 1: Estimated Asymptotic Validity Range of ρ for Percentile C.I.

Level $(1-\alpha)$	Estimated Range of ρ for which $\inf_{h_2} \lim_{n\to\infty} P(0 \in PCI_{n,1-\alpha}) \geq 1-\alpha$
90%	$[-0.90, 0.90]$
95%	$[-0.93, 0.93]$
99%	$[-0.97, 0.97]$

Note: The validity ranges are obtained numerically by simulating the asymptotic coverage formula. The numerical procedure uses 10,000 independent draws of (Z_1, Z_2) for each (h_2, ρ) combination, and uses the ρ grid $\{0.01j : -99 \leq j \leq 99\}$ and h_2 grid $\{0.1j : -250 \leq j \leq 250\}$.

under-rejects (often called conservative) in the degenerate case when $\rho = 0$, and over rejects (and hence is invalid) when $|\rho|$ happens to be very large. The former is a major concern in the literature on mediation analysis since the under-rejection means the test may have lower power. See, e.g., van Garderen and van Giersbergen (2024b).

In this section, we use the formula of the limit coverage probability derived in the previous section to design a refined PCI that under-rejects H_0 less.

Recall that in (17), we show that when $\rho = h_1 = h_2 = 0$, $\lim_{n\to\infty} \Pr(0 \in PCI_{n,1-\alpha}) = (1-\alpha)(1-\log(1-\alpha))$. If $\rho = 0 = h_1 = h_2$ is the only case that we care about, then it is easy to design a refined PCI that has exact asymptotic coverage. We can simply let the refined PCI of level $1-\alpha$ be the PCI with nominal level $1-\beta$, where β solves

$$(1-\beta)(1-\log(1-\beta)) = 1-\alpha. \tag{18}$$

Then (17) implies that the so-defined refined PCI has asymptotic coverage probability exactly $1-\alpha$. For example, when the desired coverage probability is 95%, the refined PCI is $PCI_{n,1-\beta}$, where $1-\beta \approx 0.7$.

However, when testing H_0, we typically do not know the magnitude of θ_1 and θ_2 (and hence h_1, h_2). The refinement above is too aggressive, leading to an invalid confidence interval when nonzero values of h_2 (or h_1) are allowed.

To obtain a more robust refinement, we retain the idea of nominal level modification and design a data-dependent modification based on the conditional distribution of $F_h(0|Z)$. Our design is the easiest to explain under the assumptions $\rho = 0$ and $h_1 = 0$. In this case,

$$F_h(0|Z) = \Phi(Z_1) + \Phi(Z_2 + h_2) - 2\Phi(Z_1)\Phi(Z_2 + h_2). \tag{19}$$

Let $U = \Phi(Z_1)$ and $V_{h_2} = \Phi(Z_2 + h_2)$. Then $U \sim \text{Uniform}[0,1]$, and it is independent of V_{h_2}. Now consider

the conditional probability:

$$\begin{aligned}\Pr(F_h(0|Z) \in (\beta/2, 1-\beta/2)|Z_2) &=_d \Pr(U(1-2V_{h_2}) + V_{h_2} \in (\beta/2, 1-\beta/2)|V_{h_2}) \\ &= \begin{cases} \left(\frac{1-\beta/2-V_{h_2}}{1-2V_{h_2}} \wedge 1\right) - \left(\frac{\beta/2-V_{h_2}}{1-2V_{h_2}} \vee 0\right) & V_{h_2} < 1/2 \\ 1 & V_{h_2} = 1/2 \\ \left(\frac{\beta/2-V_{h_2}}{1-2V_{h_2}} \wedge 1\right) - \left(\frac{1-\beta/2-V_{h_2}}{1-2V_{h_2}} \vee 0\right) & V_{h_2} > 1/2 \end{cases} \\ &= \begin{cases} 1 & V_{h_2} \in (\beta/2, 1-\beta/2) \\ \frac{1-\beta}{|1-2V_{h_2}|} & \text{otherwise} \end{cases}, \end{aligned} \tag{20}$$

where $\wedge$ is the minimum operator and $\vee$ is the maximum operator. Our idea is to choose a β that makes this conditional coverage probability equal to $1-\alpha$ in the worst case. In other words, we choose β to satisfy: $1-\alpha = \frac{1-\beta}{|1-2V_{h_2}|}$. Solving this equation, we get

$$\beta = 1 - (1-\alpha)|1 - 2\Phi(Z_2 + h_2)|. \tag{21}$$

This β is infeasible, of course, because Z_2 and h_2 are not observed quantities. However, recall that $Z_2 + h_2$ is the distributional limit of $\sqrt{n}\widehat{\theta}_{2n}/\sigma_2$. This suggests that β can be estimated by

$$\widehat{\beta}_n^0 = 1 - (1-\alpha)|1 - 2\Phi(\sqrt{n}\widehat{\theta}_{2n}/\widehat{\sigma}_{2n})|, \tag{22}$$

where $\widehat{\sigma}_{2n}$ is a consistent estimator of σ_2.[6]

However, the $\widehat{\beta}_n^0$ in (22) is not the modified nominal level we use to define the refined PCI. One reason is that we do not know whether $h_1 = 0$ or $h_2 = 0$ when H_0 is true. The $\widehat{\beta}_n^0$ defined in (22) presumes $h_1 = 0$. If $h_2 = 0$ and $h_1 \neq 0$ instead, we would get $\widehat{\beta}_n^0 = 1 - (1-\alpha)|1 - 2\Phi(\sqrt{n}\widehat{\theta}_{1n}/\widehat{\sigma}_1)|$. One way to make the refinement robust to the uncertainty of the identity of the nonzero parameter is to use

$$\widehat{\beta}_n^1 = 1 - (1-\alpha)\left(|1 - 2\Phi(\sqrt{n}\widehat{\theta}_{2n}/\widehat{\sigma}_{2n})| \vee |1 - 2\Phi(\sqrt{n}\widehat{\theta}_{1n}/\widehat{\sigma}_{1n})|\right). \tag{23}$$

However, another problem arises if we use the nominal level $1 - \widehat{\beta}_n^1$ PCI as the level $1-\alpha$ refined PCI. The problem is that it can lead to unintuitive results. For example, if we observe in the data that $\sqrt{n}\widehat{\theta}_{1n}$ and $\sqrt{n}\widehat{\theta}_{2n}$ are both zero, intuitively, $H_0 : \theta_1\theta_2 = 0$ should not be rejected . However, in that case, $\widehat{\beta}_n^1 = 1$; the PCI with nominal level $1 - \widehat{\beta}_n$ is a singleton set that includes only the conditional median of $\widehat{T}_n^*$. The singleton set generically does not include 0, leading to the rejection of H_0. Unintuitive results like this do not

[6] $\widehat{\sigma}_{2n}/\sqrt{n}$ is the standard error of $\widehat{\theta}_{2n}$.

mean that the PCI with nominal level $1-\widehat{\beta}_n^1$ does not cover 0 with asymptotic coverage probability at least $1-\alpha$ (it does), but may nevertheless be considered undesirable because it rejects when the data shows little evidence of nonzero θ_1 or θ_2 from the standard t-test's perspective. See van Garderen and van Giersbergen (2024b) for a discussion of another powerful test for mediation effects with a similar undesirable property.

To mitigate the foregoing problem, we use the following nominal level for our refined percentile C.I. instead:

$$\widehat{\beta}_n(\alpha) = 1-(1-\alpha)\left((1-\alpha)\vee|1-2\Phi(\sqrt{n}\widehat{\theta}_{2n}/\widehat{\sigma}_{2n})|\vee|1-2\Phi(\sqrt{n}\widehat{\theta}_{1n}/\widehat{\sigma}_{1n})|\right). \tag{24}$$

That is, we do not allow small $|\sqrt{n}\widehat{\theta}_{1n}/\widehat{\sigma}_{1n}|$ and $|\sqrt{n}\widehat{\theta}_{2n}/\widehat{\sigma}_{2n}|$ to increase $\widehat{\beta}_n(\alpha)$ further once they drop below the standard 2-sided large sample t-test critical value.

Finally, our refined PCI of level $1-\alpha$ is defined as

$$rPCI_{n,1-\alpha} = \left[\widehat{Q}_{\widehat{T}_n^*}\left(\frac{\widehat{\beta}_n(\alpha)}{2}\right), \widehat{Q}_{\widehat{T}_n^*}\left(1-\frac{\widehat{\beta}_n(\alpha)}{2}\right)\right]. \tag{25}$$

It is the PCI with the adaptive nominal level $1-\widehat{\beta}_n(\alpha)$. The refined PCI is always a subset of the original PCI because $\widehat{\beta}_n(\alpha)\geq\alpha$ almost surely. The improvement does not disappear as $n\to\infty$ because $\widehat{\beta}_n(\alpha)$ converges to a limit level that remains strictly greater than α almost surely when both local parameters are finite.

The intuition for the asymptotic validity of the refined PCI test when $\rho=0$ is clear from the foregoing discussion. Nevertheless, we formally prove it in Theorem 3. Here we use the convention: $\Phi_2(x,-\infty,\rho)=\Phi_2(-\infty,x,\rho)=0$ and $\Phi_2(x,\infty,\rho)=\Phi_2(\infty,x,\rho)=\Phi(x)$. Therefore, $F_h(0|Z)=\Phi(Z_1)+1-2\Phi(Z_1)$ when $h_1=0$ and $h_2=\infty$, and $F_h(0|Z)=\Phi(Z_1)$ when $h_1=0$ and $h_2=-\infty$.

Theorem 3. *Consider a sequence of data generating processes along which $\sqrt{n}\theta_{1n}/\sigma_1\to h_1\in[-\infty,\infty]$, $\sqrt{n}\theta_{2n}/\sigma_2\to h_2\in[-\infty,\infty]$, at least one of h_1, h_2 is zero, and Assumption 1 holds. Also assume that $\widehat{\sigma}_{jn}\to_p\sigma_j$ for $j=1,2$. Consider $\alpha\in(0,1)$. Then*

(a) $\lim_{n\to\infty}\Pr(0\in rPCI_{n,1-\alpha})=\Pr(F_h(0|Z)\in[\beta(\alpha)/2,1-\beta(\alpha)/2])$, *where*

$$\beta(\alpha)=1-(1-\alpha)\left((1-\alpha)\vee|1-2\Phi(Z_2+h_2)|\vee|1-2\Phi(Z_1+h_1)|\right). \tag{26}$$

(b) *If $\rho=0$, $\{F_h(0|Z)\in[\beta(\alpha)/2,1-\beta(\alpha)/2]\}=\{(|Z_1+h_1|\wedge|Z_2+h_2|)\leq z_{\alpha/2}\}$ where $z_{\alpha/2}$ is the $100(1-\alpha/2)\%$ quantile of the standard normal distribution. It follows that $\lim_{n\to\infty}\Pr(0\in rPCI_{n,1-\alpha})\geq 1-\alpha$ when $\rho=0$.*

Remark 1. *The first result in Part* (b) *implies that when* $\rho = 0$*, the refined PCI is asymptotically equivalent to a simple test used in the literature (e.g. Yzerbyt et al. (2018)): the joint significance (JS) test, which rejects* H_0 *when both* θ_1 *and* θ_2 *are statistically significant based on the two-sided large sample t-test.*

Since the rejection region of the PCI is a subset of that of the rPCI, the equivalence suggests that the PCI rejection region approaches a subset of that of the JS test. This is an analogous result to Proposition 2.3 *of Miller et al. (2026) which concerns tests of* $H_0 : \theta_1\theta_2 \geq 0$*. They show that the rejection region of Kowalski (2023)'s bootstrap test (under the assumption of perfect bootstrap approximation) is a subset of that of their recommended test, the latter being a one-sided version of the JS test.*

The rPCI based test and the JS test are not equivalent when $\rho \neq 0$*. The JS test remains asymptotically valid at nonzero* ρ*'s, while the rPCI does not, as we turn to next.*[7]

4.1 Size Correction to Accommodate Nonzero ρ

It is shown in Section 3.4 that under-coverage occurs for the PCI when $|\rho|$ is close to one. The under coverage can only be worse for the refined PCI because it is narrower. To address this issue, we propose a simple size correction procedure based on widening the confidence interval. Specifically, we define

$$rPCI^{sc}_{n,1-\alpha} = \left[\widehat{Q}_{\widehat{T}^*_n}\left(\frac{\widehat{\beta}_n(\alpha)}{2}\right) - \frac{\widehat{\sigma}_{1n}\widehat{\sigma}_{2n}}{n} f_{1-\alpha}(\widehat{\rho}_n), \widehat{Q}_{\widehat{T}^*_n}\left(1 - \frac{\widehat{\beta}_n(\alpha)}{2}\right) + \frac{\widehat{\sigma}_{1n}\widehat{\sigma}_{2n}}{n} f_{1-\alpha}(\widehat{\rho}_n)\right], \tag{27}$$

where $f_{1-\alpha}(\widehat{\rho}_n)$ is the standardized size-correction factor, with $\widehat{\rho}_n$ being an estimator of ρ.[8]

The size correction factor $f_{1-\alpha}(\rho)$ is defined to be the smallest nonnegative solution to the inequality:

$$\inf_{h_1,h_2:|h_1h_2|=0} \Pr\left(\beta(\alpha)/2 \leq F_h(f_{1-\alpha}(\rho)|Z),\ F_h(-f_{1-\alpha}(\rho)|Z) \leq 1 - \beta(\alpha)/2\right) \geq 1 - \alpha, \tag{28}$$

where $F_h(s|Z)$ is defined in (14). We prove formally that the event in the parentheses on the left-hand side of (28) is the limit coverage event of the PCI of nominal level $1 - \beta(\alpha)$ widened by the size correction factor $f_{1-\alpha}(\rho)$. Thus, our definition ensures that the size-corrected confidence interval has a limit coverage probability at least $1 - \alpha$.

We can show that the size correction factor is continuous in ρ on $[-1, 1]$. This implies that it is uniformly bounded as well.

Lemma 1. *For* $\alpha \in (0, 1)$*,* $f_{1-\alpha}(\cdot)$ *is continuous and even on* $[-1, 1]$*.*

[7] The JS test for our hypothesis can use $z_{\alpha/2}$ as critical value regardless of ρ, while the one-sided JS test in Miller et al. (2026) needs a larger simulated critical value at nonzero ρ's. This is due to the difference between the null hypotheses and reminds us that analysis for one hypothesis does not automatically apply to a similar-looking other.

[8] We consider symmetric widening on the two ends of the interval for simplicity. It is conceivable that one can allocate the widening asymmetrically to maximize power.

For a general ρ value, the analytical solution of $f_{1-\alpha}(\rho)$ is not available. However, it can be obtained numerically. This is because $f_{1-\alpha}(\rho)$ is a known function of α, ρ and the distribution of Z. Among them, α is known (set by the researcher a priori), the distribution of Z is $N\left(\mathbf{0}, \left(\begin{smallmatrix} 1 & \rho \\ \rho & 1 \end{smallmatrix}\right)\right)$, also known up to the parameter ρ. We calculate them numerically and report the calculated values in Table 2. In practice, the researcher can simply look up the size-correction factor in the table using their estimated ρ.[9]

The table indicates that $f_{1-\alpha}(\rho)$ is not just uniformly bounded — it is quite small. The largest $f_{1-\alpha}(\rho)$ that we obtain is less than 0.19. To put this number into context, note that in the non-degenerate case (say $\theta_1 = 0$, $\theta_2 \neq 0$ is fixed), the PCI converges to the standard asymptotic confidence interval which has a width of $2z_{\alpha/2}\sigma_1\theta_2/\sqrt{n}$. For $\alpha = 5\%$, $z_{\alpha/2} = 1.96$. Using the most conservative size-correction would increase the width by approximately $100 \times 0.19/(1.96\sqrt{n}\theta_2/\sigma_2)\%$, which diminishes to zero as n increases. Yet, without this small correction, the refined PCI can severely over-reject when $|\rho|$ is large as shown in Theorem 2.

Another feature of the table is that it starts with $\rho = 0.76$. This is because for $|\rho| < 0.76$ our numerical procedure determines that $f_{1-\alpha}(\rho) = 0$ for all $\alpha \in \{0.1, 0.05, 0.01\}$. That is, no size correction is needed in this range.

We conclude this section with a theorem that establishes the validity of the size correction. All we need in addition to the conditions of Theorem 3 is that the estimator for ρ is consistent.

Theorem 4. *Consider a sequence of data generating processes along which $\sqrt{n}\theta_{1n}/\sigma_1 \to h_1 \in [-\infty, \infty]$, $\sqrt{n}\theta_{2n}/\sigma_2 \to h_2 \in [-\infty, \infty]$, at least one of h_1, h_2 is zero, and Assumption 1 holds. Also assume that $\widehat{\sigma}_{jn} > 0$ for all n, $\widehat{\sigma}_{jn} \to_p \sigma_j$ for $j = 1, 2$, $\widehat{\rho}_n \in [-1, 1]$ for all n, and $\widehat{\rho}_n \to_p \rho$. Consider $\alpha \in (0, 1)$. Then*

$$\lim_{n\to\infty} \Pr(0 \in rPCI^{sc}_{n,1-\alpha}) \geq 1 - \alpha. \tag{29}$$

For finite h the coverage limit is

$$\Pr\left\{ \frac{\beta(\alpha)}{2} \leq F_h(f_{1-\alpha}(\rho) \mid Z),\ F_h(-f_{1-\alpha}(\rho) \mid Z) \leq 1 - \frac{\beta(\alpha)}{2} \right\}. \tag{30}$$

If one local parameter is infinite, the coverage limit is $1 - \alpha$.

Remark 2. *The last line of the theorem shows that the size-corrected and refined PCI is not conservative.*

[9] Our numerical procedure is described in Supplemental Appendix D.

Table 2: Numerically computed size correction term $f_{1-\alpha}(\rho)$

$\lvert\rho\rvert$	$f_{0.9}(\rho)$	$f_{0.95}(\rho)$	$f_{0.99}(\rho)$
< 0.76	0.000	0.000	0.000
0.76	0.001	0.000	0.000
0.77	0.002	0.000	0.000
0.78	0.007	0.000	0.000
0.79	0.011	0.000	0.000
0.80	0.016	0.000	0.000
0.81	0.019	0.000	0.000
0.82	0.025	0.003	0.000
0.83	0.029	0.006	0.000
0.84	0.034	0.010	0.000
0.85	0.042	0.017	0.000
0.86	0.047	0.021	0.000
0.87	0.052	0.025	0.000
0.88	0.060	0.032	0.001
0.89	0.067	0.039	0.007
0.90	0.073	0.044	0.012
0.91	0.082	0.052	0.017
0.92	0.089	0.057	0.019
0.93	0.099	0.068	0.028
0.94	0.106	0.075	0.036
0.95	0.116	0.082	0.042
0.96	0.125	0.092	0.054
0.97	0.135	0.101	0.058
0.98	0.147	0.112	0.067
0.99	0.164	0.129	0.080
1.00	0.187	0.148	0.096

5 Monte Carlo Simulations

We conduct Monte Carlo simulations for two examples to illustrate the finite sample performance of the refined and size-corrected PCI test.

5.1 Monte Carlo Exercise 1: Product of Means

We first show the finite sample null-rejection properties of the PCI and refined PCI tests in a simple setting where θ_1 and θ_2 are population means of observables X_1 and X_2. In this simple setting, it is straightforward to consider a rich grid of ρ values on the whole interval $[-1, 1]$. That allows us to evaluate the finite sample performance of the size-correction factors that we tabulate.

Fix a (ρ, h_2) pair, we draw an i.i.d. sample $(X_{1i}, X_{2i})'$ from the normal distribution with mean θ and variance-covariance matrix $\left(\begin{smallmatrix} 1 & \rho \\ \rho & 1 \end{smallmatrix}\right)$, where $\rho \in [-1, 1]$ and

$$\theta := \left(\begin{smallmatrix} \theta_1 \\ \theta_2 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 0 \\ n^{-1/2} h_2 \end{smallmatrix}\right). \tag{31}$$

The local parameter h_2 governs the local drifting mean of X_{2i}.

Let the estimators of θ be

$$\widehat{\theta}_n := \left(\begin{smallmatrix} \widehat{\theta}_{1n} \\ \widehat{\theta}_{2n} \end{smallmatrix}\right) = \left(\begin{smallmatrix} \bar{X}_{1n} \\ \bar{X}_{2n} \end{smallmatrix}\right), \tag{32}$$

where $\bar{X}_{jn} = n^{-1} \sum_{i=1}^{n} X_{ji}$ for $j = 1, 2$. Let the variance estimator be $\hat{\sigma}_{jn}^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_{ji} - \bar{X}_{jn})^2$, for $j = 1, 2$. We estimate the correlation between $\widehat{\theta}_{1n}$ and $\widehat{\theta}_{2n}$ using the sample correlation,

$$\hat{\rho}_n = \frac{\frac{1}{n-1} \sum_{i=1}^{n} (X_{1i} - \bar{X}_{1n})(X_{2i} - \bar{X}_{2n})}{\hat{\sigma}_{1n} \hat{\sigma}_{2n}}. \tag{33}$$

For each sample $\{(X_{1i}, X_{2i}) : i = 1, \ldots, n\}$, we calculate the PCI (P), the refined PCI (RP), and the refined and size corrected PCI (sRP). We also calculate the joint significance test (JS) described in the previous section for comparison. The bootstrap procedures use $B = 500$ bootstrap samples, each of which is drawn with replacement from $\{(X_{1i}, X_{2i}) : i = 1, \ldots, n\}$.

We repeat the above calculation for $S = 50,000$ independent copies of the sample and average to obtain the rejection rate of the four tests for $H_0 : \theta_1 \theta_2 = 0$. We consider the following values for ρ and h_2: $\rho \in \{0.01j : j = -99, \ldots, 99\}$ and $h_2 \in \{0, 1, 5\}$. We obtain the rejection rates of the four tests for each value of the pair (ρ, h_2) and show the results in Figure 1.

The figure is consistent with our theoretical findings. First, all the PCI tests exhibit under-rejection when

h_2 is close to zero and $\rho = 0$. The refinement slightly reduces the under-rejection at $\rho = 0$ and $h_2 = 1$. The refinement has a more significant effect at $h_2 = 1$ and larger ρ values. At $h_2 = 5$, all tests behave similarly.

Second, under-rejection turns into over-rejection for the raw PCI test and the refined PCI test when $h_2 = 0$ and ρ moves from 0 toward both 1 and -1, as predicted by Theorem 2. The size-correction brings the rejection rates under control across all ρ values. This provides validation for the size-correction factors in Table 2.

Third, the rejection rates of the refined PCI and the size-corrected refined PCI are close to those of the JS test at $\rho = 0$, while they deviate from the latter as $|\rho|$ grows, consistent with the prediction of Theorem 3.

5.2 Monte Carlo Exercise 2: A Mediation Model

We now consider a mediation effect model. The purpose is to illustrate the finite sample performance of PCI tests in a more realistic setup. In this exercise, we also compare the PCI tests with a rich variety of tests for mediation effects in the literature. We review these methods before describing the Monte Carlo exercise in detail.

5.2.1 Other Methods in the Literature

There are a number of existing methods for testing $H_0 : \theta_1\theta_2 = 0$. We briefly review them here. First, there is another classical asymptotic test for mediation effects: the Sobel test (Sobel (1982)), besides JS test mentioned above. The Sobel test rejects the null if

$$\frac{|\widehat{\theta}_1\widehat{\theta}_2|}{\sqrt{\widehat{\theta}_2^2\hat{s}_1^2 + \widehat{\theta}_1^2\hat{s}_2^2}} > z_{\alpha/2}, \tag{34}$$

where $\hat{s}_j$, $j = 1, 2$, are the standard errors of $\widehat{\theta}_j$. Note that $|\widehat{\theta}_1\widehat{\theta}_2|/\sqrt{\widehat{\theta}_2^2\hat{s}_1^2 + \widehat{\theta}_1^2\hat{s}_2^2} > c > 0$ implies that $|\widehat{\theta}_1/\hat{s}_1| > c$ and $|\widehat{\theta}_2/\hat{s}_2| > c$. Thus, the Sobel rejection region is contained in the JS rejection region, and both methods control size. As noted in the literature, the Sobel test tends to have lower power than the JS test or bootstrap methods when both effects are close to zero. See e.g. MacKinnon et al. (2004).

To improve the power of the mediation test, several studies, including MacKinnon et al. (2004) and Fritz and MacKinnon (2007), recommended bias corrected or bias corrected and accelerated bootstrap because their Monte Carlo simulations suggest that these methods achieve substantially higher power. However, recent works by van Garderen and van Giersbergen (2024a,b) demonstrate that those tests fail to control size.

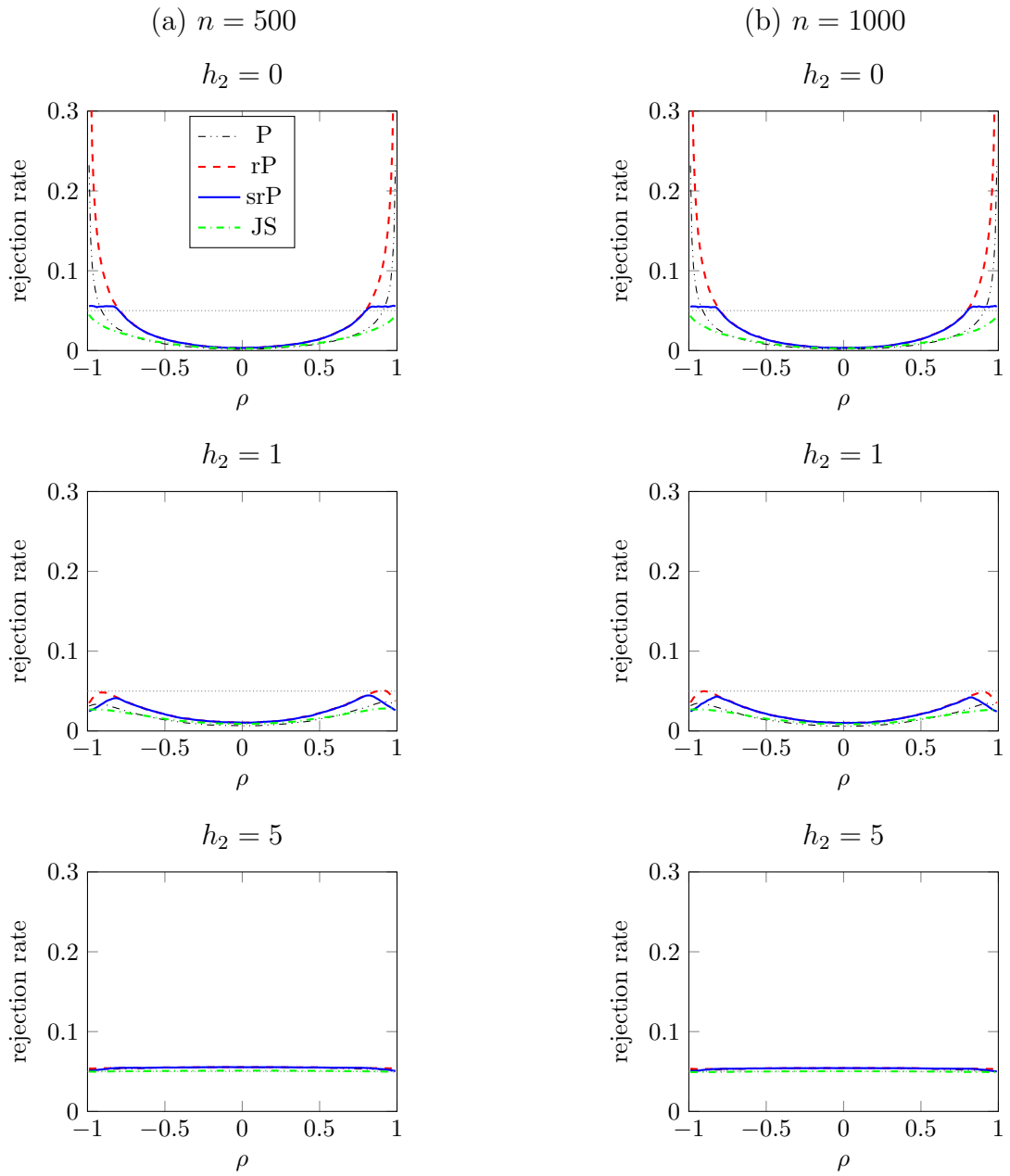


Figure 1: Rejection rates for the PCI (P), refined PCI (rP), size-corrected refined PCI (srP), and the joint significance (JS) tests in the Gaussian mean-product experiment. The panels correspond to $h_2 = 0, 1, 5$, with population means $(0, h_2/\sqrt{n})$, so the null hypothesis holds in every configuration. Each curve is based on 50,000 Monte Carlo replications with $n \in \{500, 1000\}$ and $B = 500$ bootstrap draws. The horizontal dotted line marks the nominal rejection probability of 0.05.

Recently, He et al. (2023) proposed an adaptive bootstrap method that is asymptotically pointwise exact. However, unlike PCI and our refinement, their method is motivated by pointwise asymptotic theory and is invalid when the nonzero-component of θ is close to zero. That is so even when the estimation errors are uncorrelated.

Meanwhile, van Garderen and van Giersbergen (2024b) construct a test that is optimal within the class of (nearly) similar tests for the case with $\rho = 0$. However, the test has an undesirable property of rejecting the null even when both coefficients' t-statistics are small, provided they are sufficiently close to one another. For example, a 5% test rejects when $(|t_1|, |t_2|) = (0.12, 0.11)$ or $(1.1, 1.06)$; see Table 1 of van Garderen and van Giersbergen (2024b). They do not consider cases with nonzero ρ's.

Other bootstrap methods have been used in the literature. Among them, the basic bootstrap and bootstrap-t methods represent two standard alternatives to the PCI in the bootstrap literature. The basic bootstrap rejects H_0 if $\widehat{T}_n - 0$ does not lie between the $\alpha/2$ and $(1-\alpha/2)$ bootstrap quantiles of $\widehat{T}_n^* - \widehat{T}_n$. The bootstrap-t method studentizes $\widehat{T}_n$ and $\widehat{T}_n^* - \widehat{T}_n$ with asymptotic standard errors calculated on the original sample and the bootstrap sample, respectively. They are often favored over the percentile bootstrap in the standard bootstrap asymptotic framework. However, our problem falls outside these settings, and these methods are not known to be valid.

5.2.2 The Monte Carlo Setup

For a fixed parameter value $(\theta_1, \theta_2, \beta, \gamma, a, \tau) \in \mathbb{R}^3 \times [0, \infty) \times (0, 1) \times (0, \infty)$, we generate an i.i.d. sample $\{(M_i, X_i, Y_i) : i = 1, \ldots, n\}$ according to the following model:

$$M_i = \alpha_M + \theta_1 X_i + u_i, \quad Y_i = \alpha_Y + \theta_2 M_i + \beta X_i + e_i, \tag{35}$$

where

$$u_i = \frac{W_i + \tau \varepsilon_i}{\sqrt{(1+a^2)/2 + \tau^2}}, \quad e_i = \frac{\gamma s(W_i) X_i + v_i}{\sqrt{1+\gamma^2}}, \quad s(W_i) = 2 \cdot 1\{|W_i| = 1\} - 1. \tag{36}$$

The variables X_i, v_i, ε_i are independent standard normals, and W_i is independent of them, with

$$\Pr(W_i = 1) = \Pr(W_i = -1) = \Pr(W_i = a) = \Pr(W_i = -a) = \frac{1}{4}. \tag{37}$$

In this design, the error u_i follows an equally weighted mixture of four normal distributions with means $w/\sqrt{D}$ for $w = -1, -a, a, 1$, and with common variance τ^2/D, where $D = (1+a^2)/2 + \tau^2$. The mixture is symmetric about zero and has unit variance. The random variable e_i is marginally standard normal but exhibits conditional heteroskedasticity given X_i. The construction ensures $E[X_i u_i] = E[X_i e_i] = E[u_i e_i] = 0$

and at the same time enables us to control ρ by adjusting the values of γ, a and τ. We estimate θ_1 and θ_2 by OLS. As shown in Supplemental Appendix E, the correlation of the OLS estimators $\widehat{\theta}_{1n}$ and $\widehat{\theta}_{2n}$ is

$$\rho = \frac{\gamma}{\sqrt{1+\gamma^2}} \frac{1-a^2}{1+a^2+2\tau^2}. \tag{38}$$

Hence, given $0 < |a| < 1$ and $\tau > 0$, the correlation is nonnegative when $\gamma > 0$ and increases with γ. In the simulations, we set $a = \tau = 0.2$ and $\gamma = 0, 0.3, 0.7, 5$, with the corresponding ρ being 0, 0.2463, 0.4915, 0.8405.

We consider two sample sizes $n = 500$ and $n = 1000$ and examine various parameter configurations of θ. For any bootstrap tests, the number of bootstrap repetitions is 500 throughout this section. We compute the tests in $S = 5000$ Monte Carlo repetitions and obtain the rejection rate across the 5000 repetitions for each test and each parameter specification.

We implement the following eleven methods: the PCI (P), the refined PCI (RP), the size corrected refined PCI (sRP), the bias corrected bootstrap CI (BC), the bias corrected and accelerated bootstrap CI (BCa), the basic bootstrap (B), the bootstrap-t (Bt), the adaptive bootstrap (AB) test by He et al. (2023)[10], the nearly similar test (NS) by van Garderen and van Giersbergen (2024b), the joint significance (JS) test, and Sobel's test. All the tests except for the first three and the JS test are reviewed in Section 5.2.1 above.

To implement RP, sRP, Bt, AB, NS, JS, and Sobel, we estimate the standard errors $\widehat{\sigma}_{jn}/\sqrt{n}$ of $\widehat{\theta}_{jn}$, $j = 1, 2$, using the HC1 heteroskedasticity-robust variance estimator $\widehat{\sigma}^2_{1n} = (n-2)^{-1}\sum_{i=1}^n \hat{\psi}^2_{1i}$ and $\widehat{\sigma}^2_{2n} = (n-3)^{-1}\sum_{i=1}^n \hat{\psi}^2_{2i}$, where $\hat{\psi}_{1i} = e_2'\widehat{Q}_M^{-1} Z_{Mi}\hat{\varepsilon}_{Mi}$ and $\hat{\psi}_{2i} = e_2'\widehat{Q}_Y^{-1} Z_{Yi}\hat{\varepsilon}_{Yi}$, e_2' selects the second element from the vector it is multiplied to, $\widehat{Q}_M = n^{-1}\sum_{i=1}^n Z_{Mi}Z_{Mi}'$, $\widehat{Q}_Y = n^{-1}\sum_{i=1}^n Z_{Yi}Z_{Yi}'$, $Z_{Mi} = (1, X_i)'$, $Z_{Yi} = (1, M_i, X_i)'$, and $\hat{\varepsilon}_{Mi}$ and $\hat{\varepsilon}_{Yi}$ are respectively the OLS residuals from the two regression equations in (35). The sRP additionally requires an estimate of ρ, which is used to look up the size-correction factor from Table 2. We estimate ρ using

$$\widehat{\rho}_n = \frac{\sum_{i=1}^n \hat{\psi}_{1i}\hat{\psi}_{2i}}{\sqrt{(\sum_{i=1}^n \hat{\psi}^2_{1i})(\sum_{i=1}^n \hat{\psi}^2_{2i})}}. \tag{39}$$

5.2.3 Monte Carlo Results – Size

First, we investigate the rejection rates of the test when $H_0 : \theta_1\theta_2 = 0$ is true. We consider parameter configuration $\theta_1, \theta_2 \in \{0, 2, 4, 8\}/\sqrt{n}$ satisfying $\theta_1\theta_2 = 0$.

Table 3 reports the null rejection rates at the 5% significance level, where the rejection rates larger than 0.06 are colored purple with shades deeper for larger values. As we can see, when the correlation between

[10] We chose the tuning parameter following the authors' recommendation

the coefficient estimators is weak or moderate ($\gamma = 0, 0.3$, $\rho = 0, 0.2463$), P, RP, sRP, NS, and JS all exhibit satisfactory size control across the parameter configurations considered. Sobel and B show severe under-rejection, especially when the nonzero parameter is close to zero. In contrast, BC, BCa, and Bt tend to over-reject, with rejection probabilities often exceeding 7% when the nonzero parameter is moderately large. AB performs particularly poorly away from the origin, with rejection probabilities reaching 50% or higher for several null configurations.

As the correlation increases ($\gamma = 0.7$ ($\rho = 0.4915$) and $\gamma = 5$ ($\rho = 0.8405$)), the differences across the methods become more pronounced. P and JS continue to maintain a low type I error rate. RP exhibits some over-rejection at the origin when the correlation is very high, particularly for $\gamma = 5$. sRP corrects this issue. NS exhibits noticeable size distortion at the origin, with rejection probabilities exceeding 10% when γ is large. B, BC, BCa, and Bt methods exhibit increasingly severe overrejection as the correlation increases, and AB remains highly oversized throughout. Overall, P, sRP, Sobel, and JS provide the most reliable size control across the range of correlation ($|\rho| \leq 0.8405$) structures considered in our Monte Carlo simulation.

5.2.4 Monte Carlo Results – Power

Next, we examine the power properties. Since $H_0 : \theta_1\theta_2 = 0$ is a composite hypothesis and there is no known uniformly most powerful test, we expect different tests to direct power toward different areas of the alternative space. We thus consider a variety of (θ_1, θ_2) values in the alternative space ($\theta_1\theta_2 \neq 0$), and compute the power at each of the values. Specifically, we consider θ values in the following set:

$$\Gamma_U = \left\{ \frac{(x_1, x_2)}{\sqrt{500}} : x_1, x_2 \in \{-2.5, -2.0, -1.5, 1.5, 2.0, 2.5\} \right\}.$$

The visual illustration of Γ_U is provided in Figure 2. Each point in Γ_U is numbered according to the quadrant it is in and the value of $|\theta_1\theta_2|$. The numbering is also shown in Figure 2.

We compute the power of P, sRP, JS, and Sobel, which control size well under the specifications considered, including those with high ρ. We also include NS and RP, which control size well except when ρ is large where they over-reject at $\theta = \mathbf{0}$.

Table 4 reports the weighted average powers with a uniform weight on the 36 grid points. As we can see, sRP and JS have nearly identical power when $\gamma = 0$ ($\rho = 0$), consistent with our theoretical findings. Both uniformly dominate P, while Sobel's test is substantially less powerful. sRP coincides with RP in this setting because no correction is required when the correlation is zero. The results are similar when the correlation is not large ($\gamma = 0.3$, $\rho = 0.2463$). NS has the highest weighted average power at low correlations. This advantage is unsurprising given that the test is designed to be optimal within the class of nearly similar

Table 3: Null rejection probabilities of 11 tests at the 5% significance level.

	$n = 500$							$n = 1000$						
$\sqrt{n}\theta_1$	0	0			2	4	8	0	0			2	4	8
$\sqrt{n}\theta_2$	0	2	4	8	0			0	2	4	8	0		
	$\gamma = 0$	$(\rho = 0)$												
P	0.003	0.023	0.054	0.058	0.021	0.048	0.050	0.001	0.021	0.050	0.053	0.020	0.049	0.051
RP	0.004	0.032	0.057	0.058	0.027	0.050	0.050	0.002	0.029	0.052	0.053	0.026	0.050	0.051
sRP	0.004	0.032	0.057	0.058	0.027	0.050	0.050	0.002	0.029	0.052	0.053	0.026	0.050	0.051
BC	0.007	0.053	**0.084**	**0.065**	0.045	**0.072**	**0.060**	0.005	0.049	**0.080**	**0.065**	0.050	**0.079**	**0.064**
BCa	0.007	0.052	**0.082**	**0.064**	0.045	**0.072**	**0.060**	0.004	0.048	**0.079**	**0.065**	0.050	**0.079**	**0.063**
B	0.000	0.001	0.009	0.036	0.001	0.010	0.032	0.000	0.001	0.010	0.034	0.001	0.009	0.034
Bt	0.003	0.040	**0.081**	**0.074**	0.037	**0.072**	**0.062**	0.003	0.037	**0.078**	**0.068**	0.038	**0.079**	**0.069**
AB	0.052	**0.276**	**0.519**	0.055	**0.265**	**0.504**	0.048	0.051	**0.280**	**0.559**	**0.329**	**0.276**	**0.562**	**0.343**
NS	0.053	0.053	0.057	0.056	0.052	0.050	0.050	0.045	0.051	0.054	0.054	0.047	0.052	0.051
JS	0.004	0.031	0.055	0.056	0.027	0.049	0.050	0.002	0.026	0.053	0.054	0.024	0.050	0.051
Sobel	0.000	0.003	0.025	0.046	0.004	0.023	0.043	0.000	0.003	0.023	0.045	0.004	0.022	0.043
	$\gamma = 0.3$	$(\rho = 0.2463)$												
P	0.003	0.026	0.054	0.058	0.022	0.045	0.048	0.002	0.024	0.050	0.053	0.023	0.052	0.055
RP	0.006	0.033	0.057	0.058	0.028	0.048	0.048	0.004	0.031	0.051	0.053	0.029	0.054	0.055
sRP	0.006	0.033	0.057	0.058	0.028	0.048	0.048	0.004	0.031	0.051	0.053	0.029	0.054	0.055
BC	0.007	0.050	**0.085**	**0.066**	0.048	**0.075**	0.055	0.006	0.050	**0.081**	**0.064**	0.051	**0.077**	**0.060**
BCa	0.007	0.049	**0.083**	**0.064**	0.048	**0.075**	0.055	0.006	0.050	**0.081**	**0.063**	0.051	**0.077**	**0.060**
B	0.000	0.001	0.013	0.041	0.001	0.015	0.039	0.000	0.001	0.013	0.038	0.001	0.012	0.036
Bt	0.003	0.045	**0.088**	**0.072**	0.045	**0.084**	**0.065**	0.003	0.039	**0.083**	**0.068**	0.041	**0.080**	**0.064**
AB	0.055	**0.284**	**0.532**	**0.071**	**0.285**	**0.505**	**0.065**	0.051	**0.291**	**0.578**	**0.352**	**0.286**	**0.577**	**0.354**
NS	0.057	0.058	0.056	0.056	0.050	0.049	0.049	0.051	0.053	0.054	0.054	0.052	0.052	0.052
JS	0.005	0.031	0.055	0.056	0.027	0.048	0.049	0.004	0.029	0.053	0.054	0.025	0.050	0.052
Sobel	0.000	0.006	0.024	0.047	0.006	0.022	0.043	0.000	0.006	0.022	0.044	0.004	0.023	0.044
	$\gamma = 0.7$	$(\rho = 0.4915)$												
P	0.009	0.028	0.050	0.058	0.025	0.048	0.055	0.007	0.025	0.046	0.053	0.026	0.046	0.053
RP	0.015	0.034	0.054	0.058	0.029	0.051	0.055	0.013	0.031	0.049	0.053	0.030	0.049	0.053
sRP	0.015	0.034	0.054	0.058	0.029	0.051	0.055	0.013	0.031	0.049	0.053	0.030	0.049	0.053
BC	0.015	0.052	**0.089**	**0.066**	0.052	**0.088**	**0.061**	0.013	0.050	**0.088**	**0.063**	0.047	**0.080**	**0.062**
BCa	0.015	0.049	**0.089**	**0.065**	0.051	**0.087**	**0.061**	0.013	0.050	**0.087**	**0.062**	0.047	**0.080**	**0.062**
B	0.000	0.003	0.033	0.058	0.005	0.037	**0.060**	0.000	0.004	0.033	0.057	0.003	0.033	0.051
Bt	0.004	0.057	**0.106**	**0.074**	**0.062**	**0.106**	**0.073**	0.004	**0.061**	**0.103**	**0.065**	**0.060**	**0.095**	**0.062**
AB	0.057	**0.331**	**0.545**	**0.115**	**0.325**	**0.514**	**0.115**	0.054	**0.338**	**0.609**	**0.388**	**0.327**	**0.612**	**0.392**
NS	**0.068**	0.051	0.055	0.056	0.048	0.051	0.054	**0.063**	0.047	0.051	0.054	0.053	0.050	0.053
JS	0.012	0.033	0.053	0.056	0.029	0.050	0.054	0.010	0.029	0.049	0.054	0.027	0.048	0.053
Sobel	0.001	0.010	0.025	0.048	0.010	0.022	0.045	0.001	0.008	0.021	0.045	0.006	0.021	0.045
	$\gamma = 5$	$(\rho = 0.8405)$												
P	0.029	0.034	0.042	0.058	0.028	0.039	0.054	0.029	0.028	0.035	0.053	0.026	0.033	0.051
RP	**0.064**	0.037	0.047	0.058	0.029	0.045	0.054	0.059	0.030	0.039	0.053	0.027	0.040	0.051
sRP	0.059	0.036	0.047	0.058	0.028	0.045	0.054	0.054	0.029	0.038	0.052	0.027	0.040	0.051
BC	0.035	**0.072**	**0.118**	**0.066**	**0.078**	**0.123**	**0.062**	0.035	**0.072**	**0.115**	**0.062**	**0.068**	**0.113**	0.059
BCa	0.033	**0.070**	**0.116**	**0.066**	**0.078**	**0.122**	**0.062**	0.033	**0.071**	**0.114**	**0.061**	**0.068**	**0.113**	0.058
B	0.000	**0.061**	**0.150**	**0.102**	**0.072**	**0.167**	**0.103**	0.000	**0.074**	**0.169**	**0.106**	**0.062**	**0.156**	**0.101**
Bt	0.006	**0.136**	**0.133**	**0.068**	**0.151**	**0.146**	**0.066**	0.008	**0.144**	**0.135**	0.058	**0.142**	**0.134**	0.058
AB	0.053	**0.500**	**0.557**	**0.200**	**0.502**	**0.520**	**0.196**	0.053	**0.505**	**0.655**	**0.472**	**0.492**	**0.646**	**0.484**
NS	**0.104**	0.047	0.052	0.056	0.043	0.046	0.053	**0.111**	0.041	0.044	0.054	0.037	0.040	0.049
JS	0.024	0.034	0.047	0.056	0.026	0.043	0.053	0.025	0.028	0.041	0.054	0.025	0.037	0.049
Sobel	0.003	0.014	0.025	0.048	0.012	0.019	0.042	0.004	0.012	0.019	0.044	0.012	0.019	0.041

Note: The reported values of ρ are the asymptotic correlations between the coefficient estimators, corresponding to $\gamma = 0, 0.3, 0.7, 5$, respectively.

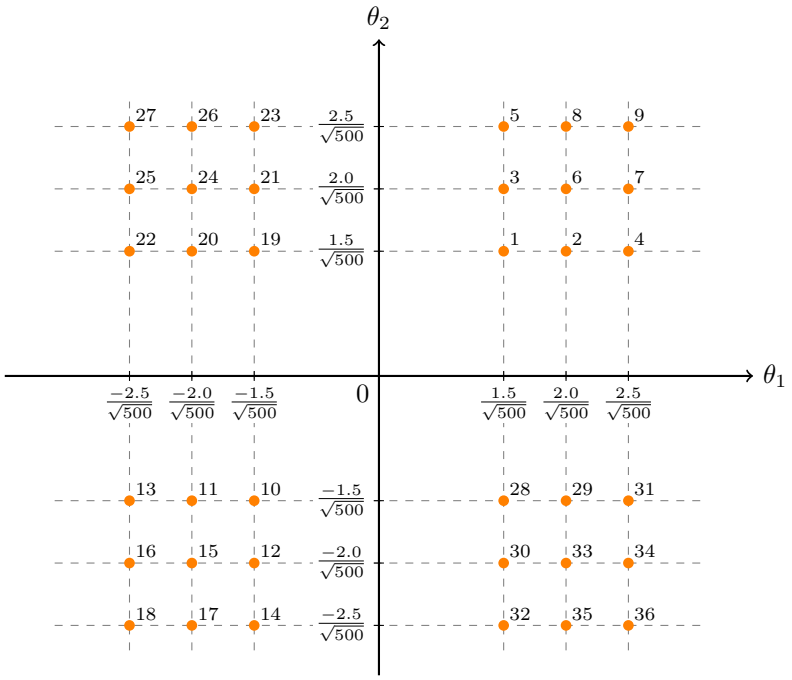


Figure 2: Illustration of Γ_U.

tests. Geometrically, the power gain of NS comes from allowing rejection when the t-statistics for both θ_1 and θ_2 are close to each other, even when both are as small as 0.11; see Table 1 in van Garderen and van Giersbergen (2024b).

As the correlation increases ($\gamma = 0.7, 5$), the power advantage of NS over sRP diminishes. They have very similar power at $\gamma = 5$ ($\rho = 0.8405$) – the over-rejection of NS in high correlation cases does not seem to lead to higher power. Also, at higher γ, the weighted average power of sRP becomes noticeably larger than that of JS. The Sobel test and the un-refined PCI continues to have lower power at high γ's.

Table 4: Weighted average power of 6 tests at the 5% significance level

		$n = 500$				$n = 1000$			
Support	Method	$\gamma = 0$	$\gamma = 0.3$	$\gamma = 0.7$	$\gamma = 5$	$\gamma = 0$	$\gamma = 0.3$	$\gamma = 0.7$	$\gamma = 5$
Γ_U	P	0.235	0.235	0.236	0.249	0.574	0.575	0.581	0.600
	RP	0.264	0.264	0.271	0.300	0.593	0.595	0.603	0.630
	sRP	0.264	0.264	0.271	0.293	0.593	0.595	0.603	0.625
	NS	0.290	0.291	0.293	0.297	0.604	0.607	0.611	0.622
	JS	0.266	0.266	0.266	0.266	0.595	0.598	0.601	0.612
	Sobel	0.108	0.106	0.104	0.105	0.409	0.408	0.407	0.402

Note: Weighted average power is computed under the uniform distribution over the alternative grid. The values $\gamma = 0, 0.3, 0.7, 5$ imply asymptotic correlations $\rho = 0, 0.2463, 0.4915, 0.8405$, respectively.

The weighted average power may mask the power tradeoff between tests. We thus also report the

rejection rates at each alternative point in Figure 2 in Figures 3 and 4. The horizontal axes show the numerical labels of the 36 points given in Figure 2. These figures show that, within each quadrant, power is largely monotonically increasing in $|\theta_1\theta_2|$ for all the tests, although power is not necessarily the same for points with the same $|\theta_1\theta_2|$, especially at high γ. Increasing the sample size increases the power for all tests, as expected. Moreover, in cases with positive correlation, all the tests have higher power in the first and third quadrants (points 19 and 10-18) than in the second and fourth quadrants (points 19-27 and 28-36). This pattern reflects the interaction between the positive correlation (ρ) and the signs of θ_1 and θ_2. When θ_1 and θ_2 have opposite signs, the positive correlation of their estimators tends to push at least one of them toward zero, making the product close to zero and hence negatively affecting the power of the tests.

These figures confirm the findings of Table 4 about the relative performance of the tests: the NS test has the highest power except at the highest correlation $\rho = 0.8405$. The sRP and the JS test are not far behind, while the Sobel test has the lowest power at every point. The power of sRP is very close to that of JS at low correlation, and it shows a noticeable power advantage over JS at high correlation against alternatives where θ_1 and θ_2 have the same signs (that is, at points labeled 1-9 and 10-18). Lastly, across all figures, one can see that sRP has a steady, uniform power advantage over P. Recall that NS and P do not control size at high correlations and that NS has the unintuitive feature of rejecting $H_0 : \theta_1\theta_2 = 0$ when the t-statistics for both θ_1 and θ_2 are very small. Given these, we recommend sRP in practice. JS is also a stronger contender since it is simple to use and has similar behavior to sRP at zero to moderate correlations.

6 Empirical Illustration

To illustrate the use of the refined and the size-corrected refined PCI, we revisit the empirical application on the Job Search Intervention Study (JOBS II) by Imai et al. (2010). JOBS II is a randomized experiment designed to evaluate the effectiveness of a job training intervention for unemployed individuals. In the experiment, the treated group participated in active-learning workshops that teach participants job-search skills and strategies for handling potential barriers and setbacks in the job-search process. Participants in the control group received informational booklets instead. Earlier studies, such as Vinokur and Schul (1997), hypothesized that this learning process would enhance job-search self-efficacy, which would in turn increase reemployment and reduce depressive symptoms.

Following this hypothesis, we define workshop participation (X) as the treatment and job-search self-efficacy as the mediator (M). The job search self-efficacy is measured as a continuous variable. The outcome (Y) is reemployment, defined as working more than 20 hours per week six months after the intervention. While Imai et al. (2010) also analyze post-treatment depression as an outcome, this paper focuses only on

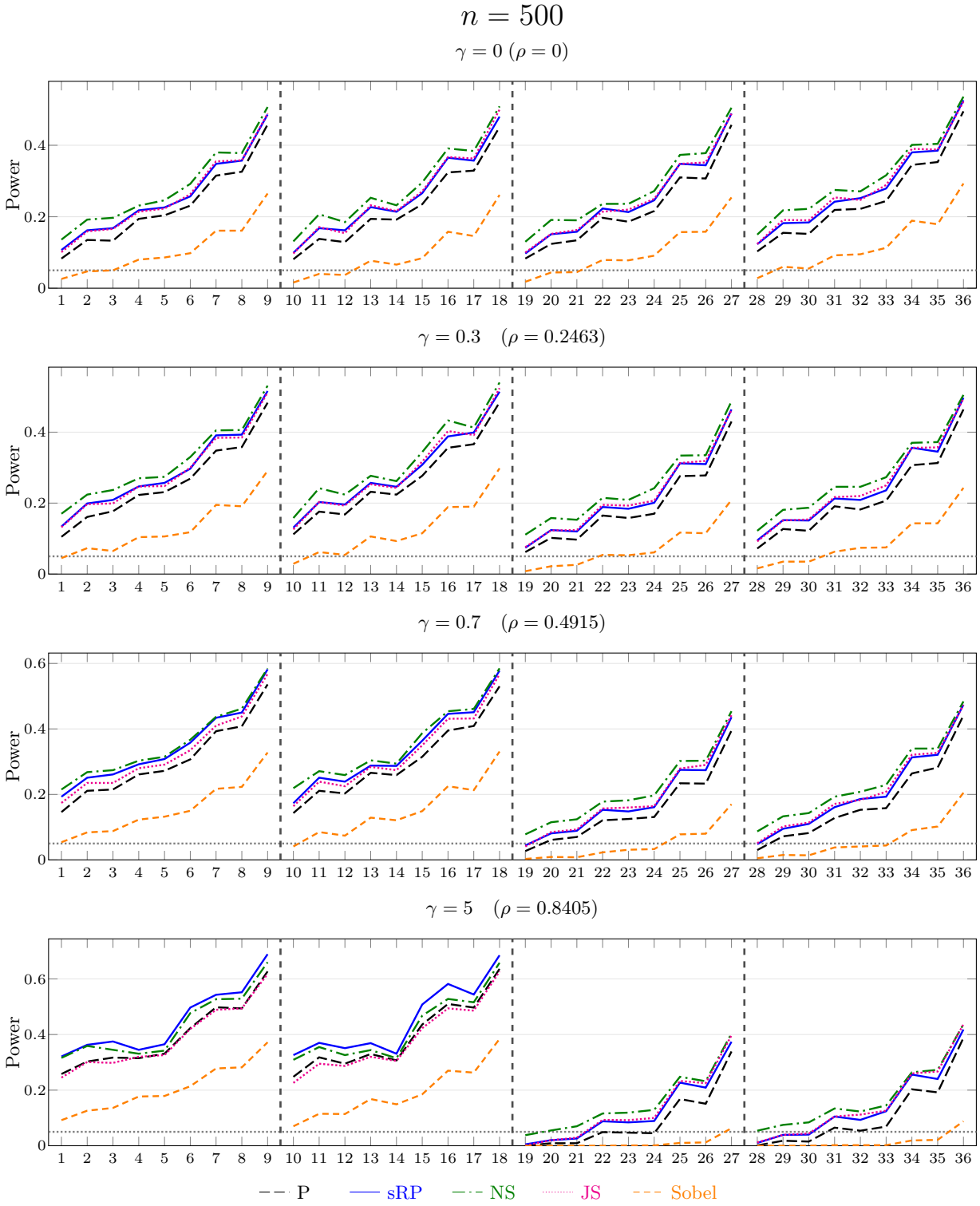


Figure 3: Power at the 36 alternatives for $n = 500$. From top to bottom, rows correspond to $\gamma = 0, 0.3, 0.7, 5$, with implied asymptotic correlations $\rho = 0, 0.2463, 0.4915, 0.8405$, respectively. Point labels follow Figure 2: labels 1–9, 10–18, 19–27, and 28–36 identify points in quadrants I, III, II, and IV, respectively. Vertical dashed lines separate quadrants. The dotted horizontal line marks the nominal level of 0.05.

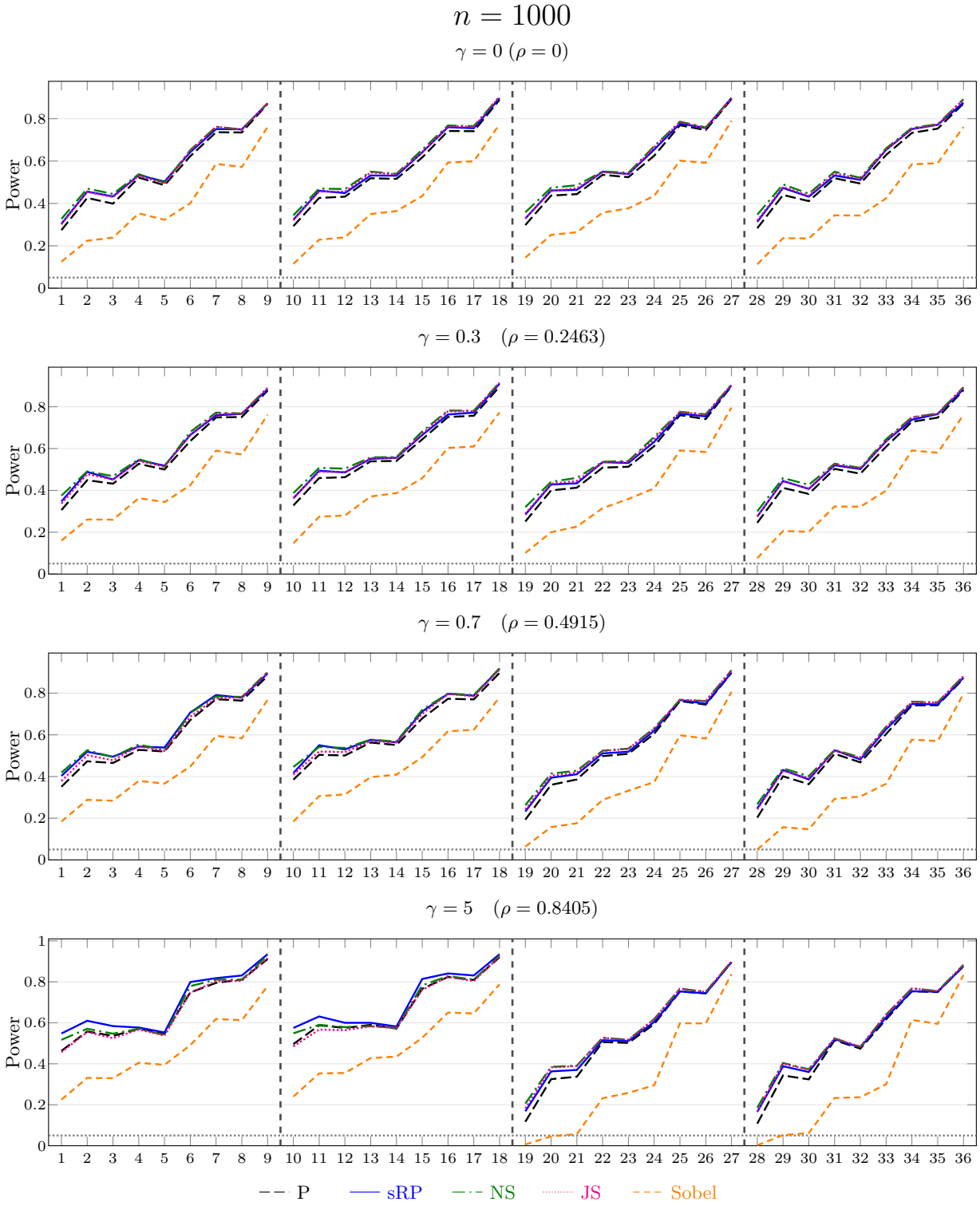


Figure 4: Power at the 36 alternatives for $n = 1000$. From top to bottom, rows correspond to $\gamma = 0, 0.3, 0.7, 5$, with implied asymptotic correlations $\rho = 0, 0.2463, 0.4915, 0.8405$, respectively. Point labels follow Figure 2: labels 1–9, 10–18, 19–27, and 28–36 identify points in quadrants I, III, II, and IV, respectively. Vertical dashed lines separate quadrants. The dotted horizontal line marks the nominal level of 0.05.

reemployment. The JOBS II study collected baseline covariates (W) prior to the treatment intervention, which include measures of education, income, race, marital status, age, sex, previous occupation, and the level of economic hardship. Using these variables as controls, we estimate the model given by (2) and (3). We estimate both equations by OLS for simplicity, even though reemployment is a binary outcome. Thus, our outcome equation (3) is estimated as a linear probability model (LPM).

Our refined PCI test proposed in Section 4 requires the standard errors of $\widehat{\theta}_{1n}$ and $\widehat{\theta}_{2n}$, and the size corrected refined PCI test requires a correlation estimator $\widehat{\rho}_n$. We calculate the standard errors using the HC1 heteroskedasticity-robust method following Imai et al. (2010). We estimate the correlation using the influence-function-based estimator $\widehat{\rho}_n$ defined in (39).

Table 5 reports the estimates and t-statistics of the individual coefficient parameters, together with their estimated correlation. Both coefficients are positive. The estimate of θ_1 is statistically significant at the 10% level, with $t_1 = 1.6608 > 1.645$, while the estimate of θ_2 is statistically significant at the 5% level, with $t_2 = 1.9602 > 1.960$. Therefore, the JS test rejects the null hypothesis of no mediation at the 10% level, but not at the 5% level. The estimated correlation is 0.0977, and hence the size correction term is zero and the refined PCI coincides with the size-corrected refined PCI.

Table 5: Estimates of the component parameters and their correlation.

Parameter	Estimate	t-statistic
θ_1	0.0798	1.6608
θ_2	0.0468	1.9602
ρ	0.0977	-

Table 6 reports the estimated mediation effect and corresponding C.I.s from this paper and Imai et al. (2010); see Table 5 of Imai et al. (2010). The estimated mediation effect in this paper is 0.0037, implying that the job training intervention increased the probability of reemployment by approximately 0.37 percentage points through its effect on job-search self-efficacy. Our inference results show that the mediation effect is significant at the 10% level using the refined PCI, but not using the conventional PCI or Sobel's method. Table 6 illustrates how the refinement in rPCI can affect inference in practice.

It is interesting to note that our estimated mediation effect matches that of Imai et al. (2010) to four decimal places, even though we use the LPM specification, while they use a probit outcome model and obtain the mediation effect by simulation. The percentile confidence intervals are also very close across the two model specifications.

Table 6: Mediation effect estimates and confidence intervals.

Estimate ($\widehat{\theta}_{1n}\widehat{\theta}_{2n}$)	Method	90% CI	95% CI
This paper			
0.0037	Percentile	$[-0.0002,\ 0.0101]$	$[-0.0008,\ 0.0119]$
	Refined percentile	$[0.0001,\ 0.0092]$	$[-0.0002,\ 0.0102]$
	Asymptotic (Sobel)	$[-0.0011,\ 0.0086]$	$[-0.0020,\ 0.0095]$
Imai et al. (2010)			
0.0037	Percentile	$[-0.0002,\ 0.0101]$	$[-0.0007,\ 0.0116]$

Note: Standard errors for asymptotic C.I.s are obtained by HC1 heteroskedasticity-robust method. Bootstrap C.I.s are based on 10,000 resamples. The test decisions are unchanged across different random seeds we used.

7 Conclusion

This paper studies the bootstrap percentile method for inference on a product parameter. We develop a theoretical framework that expresses its coverage probability directly in terms of the law of a conditional distribution function, departing from conventional analyses in the bootstrap literature. Our analysis establishes that the bootstrap percentile confidence interval is valid when the coefficient estimators are uncorrelated and remains valid over a broad range of correlations. These results provide a theoretical justification for the popular empirical use of the PCI in mediation analysis.

Our theoretical analysis further motivates a simple refinement and correction of the PCI that improves power at low to moderate correlations while preserving size control across all possible correlation statuses of the estimators of the individual parameters. Monte Carlo simulations support the theoretical findings and show that the refined and size-corrected PCI compares favorably with existing alternatives. An empirical application of job training data illustrates the performance on real data.

Our theoretical framework may be extended to other non-standard problems, such as inference about ratios of parameters and one-sided inference about products and ratios.

References

Barfield, R., Shen, J., Just, A. C., Vokonas, P. S., Schwartz, J., Baccarelli, A. A., and Lin, X. (2017). Testing for the indirect effect under the null for genome-wide mediation analyses. *Genetic Epidemiology*, 41(8):824–833.

Baron, R. M. and Kenny, D. A. (1986). The moderator-mediator variable distinction in social psycho-

logical research: Conceptual, strategic, and statistical considerations. *Journal of Personality and Social Psychology*, 51(6):1173–1182.

Cavaliere, G. and Georgiev, I. (2020). Inference under random limit bootstrap measures. *Econometrica*, 88(6):2547–2574.

Efron, B. (1979). Bootstrap Methods: Another Look at the Jackknife. *The Annals of Statistics*, 7(1):1 – 26.

Efron, B. and Tibshirani, R. J. (1994). *An Introduction to the Bootstrap*, volume 57 of *Monographs on Statistics and Applied Probability*. Chapman and Hall, New York.

Fritz, M. S. and MacKinnon, D. P. (2007). Required sample size to detect the mediated effect. *Psychological Science*, 18(3):233–239.

Hall, P. (1988). Theoretical Comparison of Bootstrap Confidence Intervals. *The Annals of Statistics*, 16(3):927 – 953.

Hall, P. (1992). *The Bootstrap and Edgeworth Expansion*. Springer Series in Statistics. Springer New York, NY.

Hansen, B. E. (2022). *Econometrics*. Princeton University Press, Princeton, NJ.

Hayes, A. F. (2009). Beyond baron and kenny: Statistical mediation analysis in the new millennium. *Communication Monographs*, 76(4):408–420.

Hayes, A. F. and Rockwood, N. J. (2017). Regression-based statistical mediation and moderation analysis in clinical research: Observations, recommendations, and implementation. *Behavior Research and Therapy*, 98:39–57.

He, Y., Song, P. X. K., and Xu, G. (2023). Adaptive bootstrap tests for composite null hypotheses in the mediation pathway analysis. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 86(2):411–434.

Hesterberg, T. C. (2015). What teachers should know about the bootstrap: Resampling in the undergraduate statistics curriculum. *The American Statistician*, 69(4):371–386.

Horowitz, J. L. (2001). Chapter 52 - the bootstrap. In Heckman, J. J. and Leamer, E., editors, *Handbook of Econometrics*, volume 5, pages 3159–3228. Elsevier.

Huber, M. (2020). *Mediation Analysis*, pages 1–38. Springer International Publishing, Cham.

Imai, K., Keele, L., and Yamamoto, T. (2010). Identification, Inference and Sensitivity Analysis for Causal Mediation Effects. *Statistical Science*, 25(1):51 – 71.

Kowalski, A. E. (2023). Behaviour within a clinical trial and implications for mammography guidelines. *The Review of Economic Studies*, 90(1):432–462.

MacKinnon, D. P., Lockwood, C. M., and Williams, J. (2004). Confidence limits for the indirect effect: Distribution of the product and resampling methods. *Multivariate Behavioral Research*, 39(1):99–128.

Miller, D. L., Molinari, F., and Stoye, J. (2026). Testing sign congruence between two parameters. *Journal of Applied Econometrics*, 41(1):3–11.

Sobel, M. E. (1982). Asymptotic confidence intervals for indirect effects in structural equation models. *Sociological Methodology*, 13:290–312.

van Garderen, K. J. and van Giersbergen, N. (2024a). Moderating the mediation bootstrap for causal inference. *Working Paper*.

van Garderen, K. J. and van Giersbergen, N. (2024b). A Nearly Similar Powerful Test for Mediation. *The Review of Economics and Statistics*, pages 1–26.

Vinokur, A. D. and Schul, Y. (1997). Mastery and inoculation against setbacks as active ingredients in the JOBS intervention for the unemployed. *Journal of Consulting and Clinical Psychology*, 65(5):867–877.

Yzerbyt, V., Muller, D., Batailler, C., and Judd, C. M. (2018). New recommendations for testing indirect effects in mediational models: The need to report and test component paths. *Journal of Personality and Social Psychology*, 115(6):929–943.

A The Asymptotic Independence Lemma

In this section, we present the lemma that proves the asymptotic independence of $\widehat{\theta}_{1n}$ and $\widehat{\theta}_{2n}$ in the correctly specified linear mediation effect model.

Lemma 2. *For the model in (2)-(3) and OLS estimators $\widehat{\theta}_{1n}$ and $\widehat{\theta}_{2n}$, additionally assume that*

(i) $E[e|W, X, M] = 0$,

(ii) $E[(1, X, W')'(1, X, W')]$ *and* $E[(1, M, X, W')'(1, M, X, W')]$ *have full rank,*

(iii) *The observations* $\{(Y_i, X_i, M_i, W_i')'\}_{i=1}^n$ *are independent and identically distributed,*

(iv) *All the observables have finite fourth moments.*

Then,

$$\sqrt{n}\begin{pmatrix}\widehat{\theta}_{1n}-\theta_1\\ \widehat{\theta}_{2n}-\theta_2\end{pmatrix} \xrightarrow{d} N\left(\mathbf{0}, \begin{pmatrix}\sigma_1^2 & 0\\ 0 & \sigma_2^2\end{pmatrix}\right), \tag{40}$$

for some $\sigma_1, \sigma_2 \geq 0$. Consequently, $\widehat{\theta}_{1n}$ and $\widehat{\theta}_{2n}$ are asymptotically independent.

Proof of Lemma 2. Let $X^{\perp}$ be the population residual from the linear projection of X on $(1, W')$, and let $M^{\perp}$ be the population residual from the linear projection of M on $(1, W')$. From (2), we get

$$M^{\perp} = \theta_1 X^{\perp} + u. \tag{41}$$

Similarly, let $Y^{\perp}_{(X)}$ and $M^{\perp}_{(X)}$ be the population residuals from the linear projection of Y and M, respectively, on $(1, X, W')$. Then from (3),

$$Y^{\perp}_{(X)} = \theta_2 M^{\perp}_{(X)} + e. \tag{42}$$

The orthogonality condition in (2) also implies that $M^{\perp}_{(X)} = u$. Thus,

$$Y^{\perp}_{(X)} = \theta_2 u + e. \tag{43}$$

Then under conditions (ii)-(iv), the Frisch–Waugh–Lovell theorem and the standard OLS expansion give

$$\sqrt{n}(\widehat{\theta}_{1n} - \theta_1) = Q_X^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^{n} X_i^{\perp}u_i + o_p(1), \text{ where } Q_X = E[(X^{\perp})^2], \tag{44}$$

$$\sqrt{n}(\widehat{\theta}_{2n} - \theta_2) = Q_M^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^{n} u_i e_i + o_p(1), \text{ where } Q_M = E[u^2]. \tag{45}$$

The covariance of the two leading summands without the scaling Q_X^{-1} and Q_M^{-1} is

$$E[X^{\perp}u^2e]. \tag{46}$$

Because $u = M - \alpha_M - \theta_1 X - \delta_M' W$, the factor $X^{\perp}u^2$ is measurable with respect to (X, M, W). Hence, by condition (i),

$$E[X^{\perp}u^2e] = E\left[X^{\perp}u^2E[e \mid X, M, W]\right] = 0. \tag{47}$$

The model's orthogonality conditions imply $E[X^{\perp}u] = E[ue] = 0$. Assumption (iv) ensures that $X^{\perp}u$ and ue have finite second moments. Thus, by the multivariate central limit theorem, the two scaled leading sums in (44) and (45) converge jointly to a centered Gaussian vector. The preceding calculation shows that the covariance of the two components of the centered Gaussian vector is zero. Standard calculations show that the variances of the two components are $\sigma_1^2 = Q_X^{-2}E[(X^{\perp})^2u^2]$ and $\sigma_2^2 = Q_M^{-2}E[u^2e^2]$.

Combining this convergence with (44)–(45) and Slutsky's theorem gives the asserted limit:

$$\sqrt{n}\begin{pmatrix}\widehat{\theta}_{1n}-\theta_1\\ \widehat{\theta}_{2n}-\theta_2\end{pmatrix} \to_d N\left(\mathbf{0}, \begin{pmatrix}\sigma_1^2 & 0\\ 0 & \sigma_2^2\end{pmatrix}\right). \tag{48}$$

The components of this Gaussian limit are independent. □

B Proofs of Formal Results in the Main Text

In this section, we present the proofs of the formal results in the main text. The proofs make use of the auxiliary lemmas in Supplemental Appendix C.

Proof of Theorem 1. It suffices to prove the case $h_1 = 0$; the other case follows by interchanging the two coordinates. By Lemma 4(a), we have

$$\begin{aligned}\left\{\widehat{F}_n^*(0) \geq \alpha/2, \widehat{F}_n^*(0-) < 1-\alpha/2\right\} &\subseteq \{0 \in PCI_{n,1-\alpha}\}\\ &\subseteq \left\{\widehat{F}_n^*(0) \geq \alpha/2, \widehat{F}_n^*(0-) \leq 1-\alpha/2\right\}.\end{aligned} \tag{49}$$

First suppose that $h_2 \in \mathbb{R}$. Then (49) and Lemmas 4(b) and 5 together give us

$$\Pr(0 \in PCI_{n,1-\alpha}) \to \Pr\left(\frac{\alpha}{2} \leq F_h(0 \mid Z) \leq 1-\frac{\alpha}{2}\right). \tag{50}$$

Let $U = \Phi(Z_1)$ and $V = \Phi(Z_2 + h_2)$. Then U is uniform on $(0,1)$ and since $\rho = 0$, U is independent of V. By (15) and using $\rho = 0$, we get

$$F_h(0 \mid Z) = U + V - 2UV = (1 - 2V)U + V. \tag{51}$$

For fixed $V = v$, the function $u \mapsto v + (1-2v)u$ is affine, with endpoint values v and $1 - v$.

If $v < \alpha/2$, then $1 - 2v > 0$ and solving both inequalities in (50) gives

$$\frac{\alpha/2 - v}{1 - 2v} \le U \le \frac{1 - \alpha/2 - v}{1 - 2v}. \tag{52}$$

Both end points lie in $(0,1)$, and the length of this interval is

$$\frac{1-\alpha}{1-2v}. \tag{53}$$

If $\alpha/2 \le v \le 1 - \alpha/2$, both endpoint values v and $1 - v$ of $u \mapsto v + (1-2v)u$ lie in $[\alpha/2, 1 - \alpha/2]$. Thus, the inequalities in (50) hold for $U \in (0,1)$. The length of this interval is 1.

Finally, if $v > 1 - \alpha/2$, then $1 - 2v < 0$ and solving the inequalities in (50), with the order reversed after division by $1 - 2v$, gives

$$\frac{1 - \alpha/2 - v}{1 - 2v} \le U \le \frac{\alpha/2 - v}{1 - 2v}, \tag{54}$$

which lies entirely in $(0,1)$ and has length

$$\frac{1-\alpha}{2v-1}. \tag{55}$$

Therefore, $\Pr\left(\frac{\alpha}{2} \le F_h(0|Z) \le 1 - \frac{\alpha}{2} \,\middle|\, V\right) = p(V)$. And hence, by the law of iterated expectations,

$$\Pr\left(\frac{\alpha}{2} \le F_h(0 \mid Z) \le 1 - \frac{\alpha}{2}\right) = E[p(V)] = E[p(\Phi(Z_2 + h_2))]. \tag{56}$$

This and (50) together show (16) when $h_2 \in \mathbb{R}$.

It remains to consider $|h_2| = \infty$.

Suppose that $h_2 = +\infty$. Then Lemma 6 implies that

$$\Pr(Z^*_{2n} + Z_{2n} \le -h_{2n} | \mathcal{F}_n) \to_p 0. \tag{57}$$

Then consider the derivation:

$$\begin{aligned}
\widehat{F}_n^*(0) &= \Pr((Z_{1n}^* + Z_{1n} + h_{1n})(Z_{2n}^* + Z_{2n} + h_{2n}) \leq 0 | \mathcal{F}_n) \\
&= \Pr((Z_{1n}^* + Z_{1n} + h_{1n})(Z_{2n}^* + Z_{2n} + h_{2n}) \leq 0, Z_{2n}^* + Z_{2n} + h_{2n} \leq 0 | \mathcal{F}_n) \\
&\quad + \Pr(Z_{1n}^* + Z_{1n} + h_{1n} \leq 0, Z_{2n}^* + Z_{2n} + h_{2n} > 0 | \mathcal{F}_n) \\
&= o_p(1) + \Pr(Z_{1n}^* + Z_{1n} + h_{1n} \leq 0 | \mathcal{F}_n) - \Pr(Z_{1n}^* + Z_{1n} + h_{1n} \leq 0, Z_{2n}^* + Z_{2n} + h_{2n} \leq 0 | \mathcal{F}_n) \\
&= \Pr(Z_{1n}^* + Z_{1n} + h_{1n} \leq 0 | \mathcal{F}_n) + o_p(1), \qquad (58)
\end{aligned}$$

where the third and the fourth equalities hold by (57). The same decomposition, with the product inequality being replaced by $<$ gives us

$$\widehat{F}_n^*(0-) = \Pr(Z_{1n}^* + Z_{1n} + h_{1n} < 0 | \mathcal{F}_n) + o_p(1). \qquad (59)$$

Since $h_{1n} \to h_1 = 0$, by the same arguments as those for Lemma 3, the conditional law of $Z_{1n}^* + Z_{1n} + h_{1n}$ given $\mathcal{F}_n$ converges in distribution to the conditional law of $Z_1^* + Z_1$ given Z_1. And by the same arguments as those for (C.17) in the proof of Lemma 4 and those for Lemma 5, we have

$$\begin{pmatrix} \Pr(Z_{1n}^* + Z_{1n} + h_{1n} \leq 0 | \mathcal{F}_n) \\ \Pr(Z_{1n}^* + Z_{1n} + h_{1n} < 0 | \mathcal{F}_n) \end{pmatrix} \to_d \begin{pmatrix} \Pr(Z_1^* + Z_1 \leq 0 | Z_1) \\ \Pr(Z_1^* + Z_1 \leq 0 | Z_1) \end{pmatrix} = \begin{pmatrix} 1 - \Phi(Z_1) \\ 1 - \Phi(Z_1) \end{pmatrix}. \qquad (60)$$

This and (58)-(59) together imply that

$$\begin{pmatrix} \widehat{F}_n^*(0) \\ \widehat{F}_n^*(0-) \end{pmatrix} \to_d \begin{pmatrix} 1 - \Phi(Z_1) \\ 1 - \Phi(Z_1) \end{pmatrix}. \qquad (61)$$

Since $\Phi(Z_1)$ has a continuous CDF, the distributional convergence and (49) together imply that

$$\Pr(0 \in PCI_{n,1-\alpha}) \to \Pr(\alpha/2 \leq 1 - \Phi(Z_1) \leq 1 - \alpha/2) = 1 - \alpha. \qquad (62)$$

On the other hand, $E[p(\Phi(Z_2 + h_2))] = p(1) = 1 - \alpha$ when $h_2 = \infty$ where we use the convention $\Phi(+\infty) = 1$. Therefore, when $h_2 = \infty$ we also have

$$\Pr(0 \in PCI_{n,1-\alpha}) \to E[p(\Phi(Z_2 + h_2))]. \qquad (63)$$

The case with $h_2 = -\infty$ follows from analogous arguments. □

Proof of Theorem 2. It suffices to prove the case $h_1 = 0$ and $h_2 \in \mathbb{R}$; the other case follows by interchanging

the two coordinates. Same as in Theorem 1,

$$\Pr(0 \in PCI_{n,1-\alpha}) \to \Pr(\alpha/2 \le F_h(0|Z) \le 1-\alpha/2). \tag{64}$$

This and (15) together show part (a).

For the limit result in part (b), note that $\alpha/2$ and $1-\alpha/2$ are continuity points of the CDFs of both $|\Phi(Z_1)-\Phi(Z_1+h_2)|$ and $1-|\Phi(Z_1)-\Phi(Z_1-h_2)|$ regardless of the value of h_2. Then it follows from Lemmas 8 and 9 that

$$\begin{aligned}
\lim_{\rho\to 1} \Pr(F_h(0|Z) \in (\alpha/2, 1-\alpha/2)) &= \Pr(|\Phi(Z_1) - \Phi(Z_1+h_2)| \in (\alpha/2, 1-\alpha/2)), \\
\lim_{\rho\to -1} \Pr(F_h(0|Z) \in (\alpha/2, 1-\alpha/2)) &= \Pr(1 - |\Phi(Z_1) - \Phi(Z_1-h_2)| \in (\alpha/2, 1-\alpha/2)) \\
&= \Pr(|\Phi(Z_1) - \Phi(Z_1-h_2)| \in (\alpha/2, 1-\alpha/2)).
\end{aligned} \tag{65}$$

Combining this with part (a) gives us the limit result in part (b).

Now we derive the zero coverage probability result in part (b). Observe that $\max_{z\in\mathbb{R}} |\Phi(z)-\Phi(z\pm h_2)| = \max_{z\in\mathbb{R}}(\Phi(z+|h_2|)-\Phi(z))$. At fixed h_2, the function $f(z) = \Phi(z+|h_2|)-\Phi(z)$ achieves its maximum at $z = -|h_2|/2$. The maximum value is $2\Phi(|h_2|/2)-1$. Therefore, $\Pr(|\Phi(Z_1)-\Phi(Z_1\pm h_2)| \in (\alpha/2, 1-\alpha/2)) = 0$ if

$$2\Phi(|h_2|/2) - 1 \le \alpha/2. \tag{66}$$

This inequality is equivalent to $|h_2| \le 2\Phi^{-1}(1/2+\alpha/4)$. Thus, the zero coverage probability result in part (b) is also proved. □

Proof of Theorem 3. First consider the case that $h \in \mathbb{R}^2$. By Lemma 4(a),

$$\begin{aligned}
\left\{\widehat{F}_n^*(0) \ge \widehat{\beta}_n(\alpha)/2, \widehat{F}_n^*(0-) < 1-\widehat{\beta}_n(\alpha)/2\right\} &\subseteq \{0 \in rPCI_{n,1-\alpha}\} \\
&\subseteq \left\{\widehat{F}_n^*(0) \ge \widehat{\beta}_n(\alpha)/2, \widehat{F}_n^*(0-) \le 1-\widehat{\beta}_n(\alpha)/2\right\}.
\end{aligned} \tag{67}$$

Similar arguments as those for Lemma 4(b) show that

$$\left(\widehat{F}_n^*(0), \widehat{F}_n^*(0-), X_n\right) \to_d \left(F_h(0|Z), F_h(0|Z), \left(\begin{smallmatrix}\sigma_1 & 0\\ 0 & \sigma_2\end{smallmatrix}\right) Z\right), \tag{68}$$

where $X_n = \begin{pmatrix} \sqrt{n}(\widehat{\theta}_{1n} - \theta_{1n}) \\ \sqrt{n}(\widehat{\theta}_{2n} - \theta_{2n}) \end{pmatrix}$. Consider

$$\widehat{\beta}_n(\alpha) = 1 - (1-\alpha)\left((1-\alpha) \vee |1 - 2\Phi(X_{1n}/\widehat{\sigma}_{1n} + \sigma_1 h_{1n}/\widehat{\sigma}_{1n})| \vee |1 - 2\Phi(X_{2n}/\widehat{\sigma}_{2n} + \sigma_2 h_{2n}/\widehat{\sigma}_{2n})|\right). \quad (69)$$

It is a continuous function of X_n, h_n, $\widehat{\sigma}_{1n}$, and $\widehat{\sigma}_{2n}$. Also, by the assumptions in the theorem, $h_n \to h$, $\widehat{\sigma}_{jn} \to_p \sigma_j$ for $j = 1, 2$. These imply that

$$\left(\widehat{F}_n^*(0), \widehat{F}_n^*(0-), X_n, h_n, \widehat{\sigma}_{1n}, \widehat{\sigma}_{2n}\right) \to_d \left(F_h(0|Z), F_h(0|Z), \left(\begin{smallmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{smallmatrix}\right) Z, h, \sigma_1, \sigma_2\right). \quad (70)$$

Thus, by the continuous mapping theorem

$$\left(\widehat{F}_n^*(0), \widehat{F}_n^*(0-), \widehat{\beta}_n(\alpha)\right) \to_d \left(F_h(0|Z), F_h(0|Z), \beta(\alpha)\right), \quad (71)$$

where $\beta(\alpha) = 1 - (1-\alpha)\left((1-\alpha) \vee |1 - 2\Phi(Z_1 + h_1)| \vee |1 - 2\Phi(Z_2 + h_2)|\right)$. Lemma 15 implies that 0 is a continuity point of the distributions of $F_h(0|Z) - \beta(\alpha)/2$ and $F_h(0|Z) - (1 - \beta(\alpha)/2)$. Then the convergence (71) implies part (a) in the case $h_1, h_2 \in \mathbb{R}$.

When $|h_1| = \infty$ or $|h_2| = \infty$, we have $\widehat{\beta}_n(\alpha) \to 1 - (1-\alpha)((1-\alpha) \vee 1) = \alpha$. Then the same arguments as those in Theorem 1 show that $\lim_{n\to\infty} \Pr(0 \in rPCI_{n,1-\alpha}) = 1 - \alpha = \Pr(F_h(0|Z) \in [\beta(\alpha)/2, 1 - \beta(\alpha)/2])$, where we use the convention $\Phi(\infty) = 1, \Phi(-\infty) = 0$, and $\Phi_2(\infty, y, \rho) = \Phi_2(y, \infty, \rho) = \Phi(y)$.

For part (b), let $U = \Phi(Z_1 + h_1)$, $V = \Phi(Z_2 + h_2)$, $\tilde{U} = 1 - 2U$ and $\tilde{V} = 1 - 2V$. Since $\rho = 0$, we have

$$\begin{aligned} F_h(0|Z) = U + V - 2UV = \frac{1 - \tilde{U}\tilde{V}}{2} \text{ and} \\ \beta(\alpha) = 1 - (1-\alpha)\left((1-\alpha) \vee |\tilde{U}| \vee |\tilde{V}|\right). \end{aligned} \quad (72)$$

Therefore, the event $F_h(0|Z) \in [\beta(\alpha)/2, 1 - \beta(\alpha)/2]$ is equivalent to

$$|\tilde{U}\tilde{V}| \le (1-\alpha)\left((1-\alpha) \vee |\tilde{U}| \vee |\tilde{V}|\right). \quad (73)$$

This is equivalent to

$$|\tilde{U}| \wedge |\tilde{V}| \le 1 - \alpha. \quad (74)$$

To see this, suppose $|\tilde{U}| \le 1 - \alpha$, then $|\tilde{U}\tilde{V}| \le (1-\alpha)|\tilde{V}| \le (1-\alpha)\left((1-\alpha) \vee |\tilde{U}| \vee |\tilde{V}|\right)$. Suppose $|\tilde{V}| \le 1 - \alpha$, then $|\tilde{U}\tilde{V}| \le (1-\alpha)|\tilde{U}| \le (1-\alpha)\left((1-\alpha) \vee |\tilde{U}| \vee |\tilde{V}|\right)$. Thus, (74) implies (73). Now suppose

that $|\tilde{U}| \geq |\tilde{V}| > 1-\alpha$. Then the right-hand side of (73) is equal to $(1-\alpha)|\tilde{U}|$, which is less than the left-hand side of (73). The same holds when $|\tilde{V}| \geq |\tilde{U}| > 1-\alpha$. Therefore, not (74) implies not (73). The two directions together implies the equivalence between (74) and (73).

Using the definition of $\tilde{U}$ and $\tilde{V}$, one can easily show that $|\tilde{U}| \leq 1-\alpha$ is equivalent to $|Z_1 + h_1| \leq z_{\alpha/2}$ and that $|\tilde{V}| \leq 1-\alpha$ is equivalent to $|Z_2 + h_2| \leq z_{\alpha/2}$. Thus, $|\tilde{U}| \wedge |\tilde{V}| \leq 1-\alpha$ is equivalent to $\min\{|Z_1 + h_1|, |Z_2 + h_2|\} \leq z_{\alpha/2}$. This shows the equality in part (b).

Since either $h_1 = 0$ or $h_2 = 0$, we have that either U or V are Uniform$[0,1]$. Suppose without loss of generality that it is U. Then

$$\Pr(|\tilde{U}| \wedge |\tilde{V}| \leq 1-\alpha) \geq \Pr(|\tilde{U}| \leq 1-\alpha) = \Pr(|1-2U| \leq 1-\alpha) = 1-\alpha, \tag{75}$$

proving the inequality in part (b). □

To prove Lemma 1, we define some new notation. For deterministic shifts (x, y), let

$$F_\rho(c; x, y) = \Pr((Z_1^* + x)(Z_2^* + y) \leq c) \tag{76}$$

for $x, y, c \in \mathbb{R}$, where Z_1^* and Z_2^* are jointly standard normal random variables with a correlation coefficient ρ. Let q denote $1-\alpha$. Let

$$\begin{aligned} Q_\rho(p; x, y) &= \inf\{c : F_\rho(c; x, y) \geq p\}, \\ b(x, y) &= \tfrac{1}{2}\big(1 - q\max\{q, |1-2\Phi(x)|, |1-2\Phi(y)|\}\big), \\ w_\rho(x, y) &= \max\{Q_\rho(b(x, y); x, y),\ -Q_\rho(1-b(x, y); x, y)\}. \end{aligned} \tag{77}$$

For the standardized conditional CDF in (14), $F_h(c \mid Z) = F_\rho(c; Z_1 + h_1, Z_2 + h_2)$ and $\beta(\alpha)/2 = b(Z_1 + h_1, Z_2 + h_2)$. In particular,

$$\alpha/2 \leq b(x, y) \leq (1-q^2)/2 < 1/2. \tag{78}$$

For the null local parameter $(0, h_2)$, set

$$W_{\rho,h_2} = w_\rho(Z_1, Z_2 + h_2), \qquad a_\alpha(\rho, h_2) = \max\{0, Q_{W_{\rho,h_2}}(q)\}, \tag{79}$$

where $Q_W(q) = \inf\{c : \Pr(W \leq c) \geq q\}$.

Proof of Lemma 1. Consider an arbitrary $(x, y) \in \mathbb{R}^2$. For every $\rho \in (-1, 1)$, $(Z_1^* + x)(Z_2^* + y)$ is a continuous

function of $(Z_1^*, Z_2^*)'$, whose density is everywhere positive on $\mathbb{R}^2$, and for $\rho = \pm 1$, $(Z_1^* + x)(Z_2^* + y) = (Z_1^* + x)(\pm Z_1^* + y)$ is a continuous function of Z_1^*, whose density is everywhere positive on $\mathbb{R}$. Therefore, by Lemma 10, for each $\rho \in [-1, 1]$, we have

$$F_\rho(s; x, y) < p \text{ if } s < Q_\rho(p; x, y), \qquad F_\rho(s; x, y) > p \text{ if } s > Q_\rho(p; x, y), \tag{80}$$

for $p \in (0, 1)$. Then by the continuity of $F_\rho(\cdot; x, y)$ proved in Lemma 11, we have

$$F_\rho(Q_\rho(p; x, y); x, y) = p. \tag{81}$$

Fix a $\rho \in [-1, 1]$ and an arbitrary candidate correction $c \geq 0$. Consider h of the shape $h = (0, h_2)$ where $h_2 \in \mathbb{R}$. Let $b = \beta(\alpha)/2 \equiv b(Z_1, Z_2 + h_2)$. Let $F(\cdot) = F_\rho(\cdot; Z_1, Z_2 + h_2)$ and $Q(\cdot) = Q_\rho(\cdot; Z_1, Z_2 + h_2)$. Then by (80) and (81), we have

$$\begin{aligned} b \leq F(c) &\iff Q(b) \leq c, \\ F(-c) \leq 1 - b &\iff -c \leq Q(1 - b) \iff -Q(1 - b) \leq c. \end{aligned} \tag{82}$$

Thus the event in (28), with $f_{1-\alpha}(\rho)$ replaced by c, is exactly $\{W_{\rho,h_2} \leq c\}$. For any real random variable W, $\Pr(W \leq c) \geq q$ if and only if $c \geq Q_W(q)$, whether or not W has atoms. Since $c \geq 0$, this gives

$$\Pr(W_{\rho,h_2} \leq c) \geq q \iff c \geq a_\alpha(\rho, h_2). \tag{83}$$

Interchanging coordinates leaves the Gaussian law and b unchanged, so the inequality in (28) holds with correction $f_{1-\alpha}(\rho) = c$ if and only if $c \geq a_\alpha(\rho, h_2)$ for every $h_2 \in \mathbb{R}$, or equivalently,

$$c \geq \sup_{h_2 \in \mathbb{R}} a_\alpha(\rho, h_2). \tag{84}$$

This supremum is finite: Lemma 14, with $\varepsilon = 1$, bounds a_α by 1 uniformly over $\rho \in [-1, 1]$ for $|h_2| \geq H_1$, while Lemma 13 bounds it on the compact set $[-1, 1] \times [-H_1, H_1]$. Thus, the supremum of $a_\alpha(\rho, h_2)$ over $h_2 \in \mathbb{R}$ is bounded uniformly over $\rho \in [-1, 1]$. Therefore,

$$f_{1-\alpha}(\rho) = \sup_{h_2 \in \mathbb{R}} a_\alpha(\rho, h_2), \tag{85}$$

and it is uniformly bounded over $\rho \in [-1, 1]$.

Lemma 12 immediately implies the evenness of $f_{1-\alpha}$.

Lemma 13 proves joint continuity of a_α on $[-1,1]\times\mathbb{R}$. For finite $H>0$, let $f_H(\rho)=\max_{|h_2|\le H} a_\alpha(\rho,h_2)$. Uniform continuity of a_α on $[-1,1]\times[-H,H]$ gives $|f_H(\rho_n)-f_H(\rho)|\le \sup_{|h_2|\le H}|a_\alpha(\rho_n,h_2)-a_\alpha(\rho,h_2)|\to 0$ as $\rho_n\to\rho$, so f_H is continuous.

Lemma 14 gives

$$\lim_{H\to\infty}\sup_{\rho\in[-1,1],\ |h_2|\ge H} a_\alpha(\rho,h_2)=0. \tag{86}$$

This implies that

$$\sup_{\rho\in[-1,1]}(f_{1-\alpha}(\rho)-f_H(\rho))\le \sup_{\rho\in[-1,1],|h_2|>H} a_\alpha(\rho,h_2)\to 0, \tag{87}$$

as $H\to\infty$. This combined with the fact that $f_{1-\alpha}(\rho)-f_H(\rho)\ge 0$ implies that $f_{1-\alpha}$ is a uniform limit of continuous functions f_H, hence continuous. □

Proof of Theorem 4. It suffices to prove the case $h_1=0$; the other case follows by interchanging the two coordinates. Let

$$\omega_n=\widehat{\sigma}_{1n}\widehat{\sigma}_{2n},\qquad f_n=f_{1-\alpha}(\widehat{\rho}_n),\qquad c_n=\frac{\omega_n f_n}{\sigma_1\sigma_2},\qquad f=f_{1-\alpha}(\rho). \tag{88}$$

By Lemma 1 and consistency of the estimators, $f_n\to_p f$ and $c_n\to_p f$.

Observe that

$$\begin{aligned}\{0\in rPCI^{sc}_{n,1-\alpha}\}&=\left\{\widehat{Q}_{\widehat{T}^*_n}\left(\frac{\widehat{\beta}_n(\alpha)}{2}\right)-\frac{\omega_n}{n}f_{1-\alpha}(\widehat{\rho}_n)\le 0\le \widehat{Q}_{\widehat{T}^*_n}\left(1-\frac{\widehat{\beta}_n(\alpha)}{2}\right)+\frac{\omega_n}{n}f_{1-\alpha}(\widehat{\rho}_n)\right\}\\&=\left\{\widehat{Q}_{\frac{n\widehat{T}^*_n}{\sigma_1\sigma_2}}\left(\frac{\widehat{\beta}_n(\alpha)}{2}\right)\le c_n,\ \widehat{Q}_{\frac{n\widehat{T}^*_n}{\sigma_1\sigma_2}}\left(1-\frac{\widehat{\beta}_n(\alpha)}{2}\right)\ge -c_n\right\}.\end{aligned} \tag{89}$$

Then by Lemma 4(a), we have

$$\left\{\begin{array}{l}\widehat{F}^*_n(c_n)\ge\widehat{\beta}_n(\alpha)/2,\\ \widehat{F}^*_n((-c_n)-)<1-\widehat{\beta}_n(\alpha)/2\end{array}\right\}\subseteq\{0\in rPCI^{sc}_{n,1-\alpha}\},$$

$$\{0\in rPCI^{sc}_{n,1-\alpha}\}\subseteq\left\{\begin{array}{l}\widehat{F}^*_n(c_n)\ge\widehat{\beta}_n(\alpha)/2,\\ \widehat{F}^*_n((-c_n)-)\le 1-\widehat{\beta}_n(\alpha)/2\end{array}\right\}. \tag{90}$$

First, suppose $h_2 \in \mathbb{R}$. Let μ_n be the conditional law of $n\widehat{T}_n^*/(\sigma_1\sigma_2)$. Lemma 3 applies and gives

$$(Z_n, \mu_n) \to_d (Z, K_h(Z)), \tag{91}$$

where $K_h(Z)$ has CDF $F_h(\cdot|Z)$. Recall that $\widehat{\beta}_n(\alpha) = 2b\left(\frac{\sqrt{n}\widehat{\theta}_{1n}}{\widehat{\sigma}_{1n}}, \frac{\sqrt{n}\widehat{\theta}_{2n}}{\widehat{\sigma}_{2n}}\right)$ and that

$$\frac{\sqrt{n}\,\widehat{\theta}_{jn}}{\widehat{\sigma}_{jn}} = \frac{\sigma_j}{\widehat{\sigma}_{jn}}(Z_{jn} + h_{jn}). \tag{92}$$

Then by the continuity of $b(\cdot,\cdot)$, the consistency of $\widehat{\sigma}_{jn}$, the convergence of h_{jn} for $j = 1, 2$, and Assumption 1(a), we have

$$(\mu_n, c_n, \widehat{\beta}_n(\alpha)) \to_d (K_h(Z), f, \beta(\alpha)). \tag{93}$$

Combining that with Lemma 4(b), using the continuous mapping theorem, gives us

$$\left(\widehat{F}_n^*(c_n), \widehat{F}_n^*((-c_n)-), \widehat{\beta}_n(\alpha)\right) \to_d \left(F_h(f|Z), F_h(-f|Z), \beta(\alpha)\right). \tag{94}$$

Lemma 15, applied at $c = f$ and $c = -f$, gives

$$\Pr\{F_h(f|Z) = \beta(\alpha)/2\} = \Pr\{F_h(-f|Z) = 1 - \beta(\alpha)/2\} = 0. \tag{95}$$

Both bounding events in (90) are therefore continuity sets for the limiting vector. Their probabilities converge to the same limit, so (90)–(94) imply

$$\begin{aligned}\lim_{n\to\infty} \Pr(0 \in rPCI^{sc}_{n,1-\alpha}) &= \Pr\left\{\frac{\beta(\alpha)}{2} \le F_h(f|Z),\ F_h(-f|Z) \le 1 - \frac{\beta(\alpha)}{2}\right\} \\ &\ge 1 - \alpha,\end{aligned} \tag{96}$$

where the inequality follows from the definition of $f_{1-\alpha}(\rho)$.

Now suppose $h_2 \in \{-\infty, \infty\}$. Then $h_1 = 0$. Hence, $h_{1n} \to 0$. Let $s = \lim h_{2n}/|h_{2n}| \in \{-1, 1\}$. Then

$$\frac{n\widehat{T}_n^*}{\sigma_1\sigma_2|h_{2n}|} = (Z_{1n}^* + Z_{1n} + h_{1n})\left(\frac{h_{2n}}{|h_{2n}|} + \frac{Z_{2n}^* + Z_{2n}}{|h_{2n}|}\right). \tag{97}$$

Let ν_n be its conditional law given $\mathcal{F}_n$. Assumption 1 gives convergence in probability of the conditional law of Z_n^* to the law of Z^*. Since the law of Z^* is deterministic, this convergence and $Z_n \to_d Z$

imply that the conditional law of Z_n^* given $\mathcal{F}_n$ and Z_n converge jointly in distribution. Then, using $(h_{1n}, h_{2n}/|h_{2n}|, 1/|h_{2n}|) \to (0, s, 0)$ and the same argument as in Lemma 3, we get

$$\nu_n \to_d \mathcal{L}\big((Z_1^* + Z_1)s \mid Z\big). \tag{98}$$

Since $c_n/|h_{2n}| \to_p 0$, applying the same arguments as those for Lemma 4(b) on the sequence of conditional laws of $n\widehat{T}_n^*/(\sigma_1\sigma_2|h_{2n}|)$ and the evaluation points $c_n/|h_{2n}|$ and $-c_n/|h_{2n}|$ yields

$$\big(\widehat{F}_n^*(c_n), \widehat{F}_n^*((-c_n)-)\big) \to_d (U_s, U_s), \qquad U_s = \Phi(-sZ_1). \tag{99}$$

The variable U_s is uniform on $(0,1)$. Also $(\sigma_2/\widehat{\sigma}_{2n})(Z_{2n} + h_{2n})$ diverges in probability with sign s, so $\widehat{\beta}_n(\alpha) \to_p \alpha$. Both bounding probabilities in (90) therefore converge to $\Pr\{\alpha/2 \le U_s \le 1 - \alpha/2\} = 1 - \alpha$. Thus, the same limit holds for $\Pr(0 \in rPCI^{sc}_{n,1-\alpha})$, completing the proof. □

Supplemental Appendix to

Testing Mediation Effects: A Theoretical Case for Bootstrap Percentile Confidence Interval

Woosik Gong Xiaoxia Shi

Appendix C states and proves the auxiliary lemmas used in Appendix B of the main paper. Appendix D describes how we compute the size-correction factors reported in the paper. Appendix E derives the correlation between the two OLS estimators in the mediation simulation.

C Auxiliary Results

In this section, we give the auxiliary lemmas that are used to prove the main results. The lemmas are also proved in this section.

Let $K_h(Z)$ denote the random probability measure defined by the conditional law of T_h (defined in (12)) given $Z = (Z_1, Z_2)'$. Let $\mathcal{P}(\mathbb{R})$ denote the space of probability measures on $\mathbb{R}$. Let the bounded-Lipschitz metric d_{BL} on $\mathcal{P}(\mathbb{R})$ be

$$d_{BL}(\mathcal{L}_1, \mathcal{L}_2) = \sup_{f \in BL_1(\mathbb{R})} \left| \int f d\mathcal{L}_1 - \int f d\mathcal{L}_2 \right|, \tag{C.1}$$

for $\mathcal{L}_1, \mathcal{L}_2 \in \mathcal{P}(\mathbb{R})$.

Lemma 3 (Convergence of conditional product law)**.** *Suppose Assumption 1 holds and* $h_n := \sqrt{n}\begin{pmatrix} \theta_{1n}/\sigma_1 \\ \theta_{2n}/\sigma_2 \end{pmatrix} \to h = \begin{pmatrix} h_1 \\ h_2 \end{pmatrix} \in \mathbb{R}^2$*. Then the conditional law of* $n\widehat{T}_n^*/(\sigma_1\sigma_2)$ *given* $\mathcal{F}_n$ *converges in distribution, as a random element in* $(\mathcal{P}(\mathbb{R}), d_{\mathrm{BL}})$*, to the limiting random kernel* $K_h(Z)$*. The convergence holds jointly with the convergence in Assumption 1*(a)*.*

Proof of Lemma 3. Let

$$X_n := \sqrt{n}\begin{pmatrix} \widehat{\theta}_{1n} - \theta_{1n} \\ \widehat{\theta}_{2n} - \theta_{2n} \end{pmatrix}, \qquad Y_n^* := \sqrt{n}\begin{pmatrix} \widehat{\theta}_{1n}^* - \widehat{\theta}_{1n} \\ \widehat{\theta}_{2n}^* - \widehat{\theta}_{2n} \end{pmatrix}. \tag{C.2}$$

Let G_n^* denote the conditional law of Y_n^* given $\mathcal{F}_n$, and let $G = N(\mathbf{0}, \Sigma)$. Assumption 1(b) gives

$$G_n^* \xrightarrow{p} G \tag{C.3}$$

in the bounded-Lipschitz metric. By Assumption 1(a) and $h_n \to h$,

$$X_n + \left(\begin{smallmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{smallmatrix}\right) h_n \to_d \begin{pmatrix} \sigma_1 Z_1 \\ \sigma_2 Z_2 \end{pmatrix} + \left(\begin{smallmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{smallmatrix}\right) h. \tag{C.4}$$

For $u \in \mathbb{R}^2$ and a probability measure ν on $\mathbb{R}^2$, define

$$\Psi(u, \nu) := \nu \circ \varphi_u^{-1}, \qquad \varphi_u(y) := \frac{(y_1 + u_1)(y_2 + u_2)}{\sigma_1 \sigma_2}. \tag{C.5}$$

Thus, $\Psi(u, \nu)$ is the law of $(Y_1 + u_1)(Y_2 + u_2)/(\sigma_1\sigma_2)$ when $Y \sim \nu$.

The map $\Psi : \mathbb{R}^2 \times \mathcal{P}(\mathbb{R}^2) \to \mathcal{P}(\mathbb{R})$ is continuous when $\mathbb{R}^2$ has its usual topology and both spaces of probability measures carry their weak topologies. To see this, suppose $u_n \to u$ and $\nu_n \Rightarrow \nu$, and let $Y_n \sim \nu_n$ and $Y \sim \nu$. Then $Y_n \to_d Y$, and because u_n is deterministic, we have $(u_n, Y_n) \to_d (u, Y)$. It then follows from the continuous mapping theorem that $\frac{(Y_{1n}+u_{1n})(Y_{2n}+u_{2n})}{\sigma_1\sigma_2} \to_d \frac{(Y_1+u_1)(Y_2+u_2)}{\sigma_1\sigma_2}$, which is equivalent to

$$\Psi(u_n, \nu_n) \Rightarrow \Psi(u, \nu). \tag{C.6}$$

Similarly, the map $\Psi^1(x, u, \nu) = (x, \Psi(u, \nu))$ is also continuous.

By (C.3) and (C.4), and because G, h_n, and h are deterministic, joint weak convergence holds for the triple:

$$\left(X_n, X_n + \left(\begin{smallmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{smallmatrix}\right) h_n, G_n^*\right) \Rightarrow \left(\left(\begin{smallmatrix} \sigma_1 Z_1 \\ \sigma_2 Z_2 \end{smallmatrix}\right), \left(\begin{smallmatrix} \sigma_1 Z_1 \\ \sigma_2 Z_2 \end{smallmatrix}\right) + \left(\begin{smallmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{smallmatrix}\right) h, G\right). \tag{C.7}$$

Applying the continuous mapping theorem to Ψ^1 gives convergence in distribution of the conditional product law:

$$\left(X_n, \Psi\left(X_n + \left(\begin{smallmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{smallmatrix}\right) h_n, G_n^*\right)\right) \Rightarrow \left(\left(\begin{smallmatrix} \sigma_1 Z_1 \\ \sigma_2 Z_2 \end{smallmatrix}\right), \Psi\left(\left(\begin{smallmatrix} \sigma_1 Z_1 \\ \sigma_2 Z_2 \end{smallmatrix}\right) + \left(\begin{smallmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{smallmatrix}\right) h, G\right)\right). \tag{C.8}$$

The second component on the left-hand side is the conditional law of $n\widehat{T}_n^*/(\sigma_1\sigma_2)$ given $\mathcal{F}_n$, since

$$\frac{n\widehat{T}_n^*}{\sigma_1\sigma_2} = \frac{n\widehat{\theta}_{1n}^*\widehat{\theta}_{2n}^*}{\sigma_1\sigma_2} = \frac{(Y_{n1}^* + X_{n1} + \sigma_1 h_{n1})(Y_{n2}^* + X_{n2} + \sigma_2 h_{n2})}{\sigma_1\sigma_2}, \tag{C.9}$$

while the second component on the right-hand side is the conditional law of $T_h = (Z_1^* + Z_1 + h_1)(Z_2^* + Z_2 + h_2)$ given Z, namely $K_h(Z)$. This proves the claims. □

Lemma 4 (Conditional percentile coverage). (a) *For any $c \in \mathbb{R}$ and $0 < p < q < 1$, and for $\widehat{F}_n^*(x) = \Pr(n\widehat{T}_n^*/(\sigma_1\sigma_2) \le x \mid \mathcal{F}_n)$ and $\widehat{F}_n^*(x-) = \Pr(n\widehat{T}_n^*/(\sigma_1\sigma_2) < x \mid \mathcal{F}_n)$, we have*

$$\left\{\widehat{F}_n^*\left(\frac{c}{\sigma_1\sigma_2}\right) \ge p, \widehat{F}_n^*\left(\left(-\frac{c}{\sigma_1\sigma_2}\right)-\right) < q\right\} \subseteq \left\{\widehat{Q}_{n\widehat{T}_n^*}(p) - c \le 0 \le \widehat{Q}_{n\widehat{T}_n^*}(q) + c\right\}$$
$$\subseteq \left\{\widehat{F}_n^*\left(\frac{c}{\sigma_1\sigma_2}\right) \ge p, \widehat{F}_n^*\left(\left(-\frac{c}{\sigma_1\sigma_2}\right)-\right) \le q\right\}. \quad \text{(C.10)}$$

(b) *Suppose that Assumption 1 holds and $(\sqrt{n}\theta_{1n}/\sigma_1, \sqrt{n}\theta_{2n}/\sigma_2)' \to h \in \mathbb{R}^2$. For sequences $c_n \to_p c$, where $c \in \mathbb{R}$ is deterministic, we have $\left(\widehat{F}_n^*(c_n), \widehat{F}_n^*((-c_n)-)\right) \to_d (F_h(c|Z), F_h(-c|Z))$.*

Proof of Lemma 4. For any distribution function F and its quantile function $Q_F(r) = \inf\{x : F(x) \ge r\}$, we have

$$Q_F(p) \le x \iff F(x) \ge p, \text{ and } x \le Q_F(q) \iff F(t) < q \text{ for all } t < x. \quad \text{(C.11)}$$

Let $F(x-) = \lim_{t\uparrow x} F(t) = \Pr(X < x)$ for $X \sim F$. Then the second equivalence implies that

$$F(x-) < q \Rightarrow x \le Q_F(q) \Rightarrow F(x-) \le q. \quad \text{(C.12)}$$

Applying these implications to $F = \widehat{F}_n^*$ and using $Q_{\widehat{F}_n^*}(r) = \widehat{Q}_{n\widehat{T}_n^*}(r)/(\sigma_1\sigma_2)$ gives part (a).

Let μ_n denote the conditional law of $n\widehat{T}_n^*/(\sigma_1\sigma_2)$ given $\mathcal{F}_n$, and let

$$\mu := K_h(Z). \quad \text{(C.13)}$$

Define maps on $\mathbb{R} \times \mathcal{P}(\mathbb{R})$ by

$$e(y,\nu) := \nu((-\infty, y]), \qquad e_-(y,\nu) := \lim_{x\uparrow y} \nu((-\infty, x]) = \nu((-\infty, y)). \quad \text{(C.14)}$$

The maps e and e_- are jointly continuous in (y,ν) at every $y \in \mathbb{R}$ and probability measure ν satisfying $\nu(\{y\}) = 0$. To see why, note that it follows from the Portmanteau theorem (see Billingsley (1999, Theorem 2.1)) that, for any $y_n \to y$ and $\nu_n \Rightarrow \nu$,

$$\nu((-\infty, y)) \le \liminf_{n\to\infty} \nu_n((-\infty, y_n)) \le \limsup_{n\to\infty} \nu_n((-\infty, y_n]) \le \nu((-\infty, y]). \quad \text{(C.15)}$$

The quantities on the two ends are equal when $\nu(\{y\}) = 0$.

Since $c_n \to_p c$ where c is deterministic and $\mu_n \Rightarrow \mu$ (by Lemma 3), we have $(\mu_n, c_n) \Rightarrow (\mu, c)$. Then, applying the continuous mapping theorem gives

$$\big(e(c_n, \mu_n), e_-(-c_n, \mu_n)\big) \Rightarrow \big(e(c, \mu), e_-(-c, \mu)\big) = \big(F_h(c \mid Z), F_h(-c \mid Z)\big), \tag{C.16}$$

where the equality holds almost surely because $\mu(\{c\}) = \mu(\{-c\}) = 0$ almost surely, by the almost sure continuity of $F_h(\cdot \mid Z)$ on $\mathbb{R}$ (Lemma 5). Since $e(c_n, \mu_n) = \widehat{F}_n^*(c_n)$ and $e_-(-c_n, \mu_n) = \widehat{F}_n^*((-c_n)-)$, the convergence in (C.16) is exactly

$$(\widehat{F}_n^*(c_n), \widehat{F}_n^*((-c_n)-)) \to_d (F_h(c|Z), F_h(-c|Z)). \tag{C.17}$$

Thus, part (b) holds. □

Lemma 5 (Verification of the boundary condition)**.** *Suppose that $\rho \in [-1, 1]$, $\sigma_1 > 0$, $\sigma_2 > 0$, and $h_1, h_2 \in \mathbb{R}$. Then, almost surely, the conditional distribution function $F_h(\cdot \mid Z)$ is continuous at c for any $c \in \mathbb{R}$. Moreover, if $\rho \in (-1, 1)$ or if $|h_1| \neq |h_2|$, the random variable $F_h(0 \mid Z)$ has no atoms.*

Proof of Lemma 5. Suppose that $\rho \in (-1, 1)$ first. Since $Z^* = (Z_1^*, Z_2^*)$ has a nonsingular bivariate normal distribution when $\rho \in (-1, 1)$, the random vector

$$(Z_1^* + Z_1 + h_1,\ Z_2^* + Z_2 + h_2) \tag{C.18}$$

has a joint density conditional on $Z = (Z_1, Z_2)$. The preimage of any singleton set of the product map $(x, y) \mapsto xy$ is either the two branches of a hyperbola or the union of the coordinate axes. In either case, it has two-dimensional Lebesgue measure zero. Hence

$$\Pr\big((Z_1^* + Z_1 + h_1)(Z_2^* + Z_2 + h_2) = c \mid Z\big) = 0 \tag{C.19}$$

almost surely. This implies that $F_h(\cdot \mid Z)$ is continuous at c for any $c \in \mathbb{R}$.

Now suppose that $\rho = \pm 1$. Then $(Z_1^* + Z_1 + h_1)(Z_2^* + Z_2 + h_2) = (Z_1^* + Z_1 + h_1)(\pm Z_1^* + Z_2 + h_2)$ is a non-constant quadratic function of Z_1^*. Thus, the pre-image of any singleton set of this mapping contains at most two values and therefore has Lebesgue measure zero. Since the distribution of Z_1^* has no atom given Z, (C.19) also holds in this case.

Therefore, for $\rho \in [-1, 1]$, we have that $F_h(\cdot|Z)$ is continuous at c for any $c \in \mathbb{R}$, almost surely.

By definition, $T_h = (Z_1^* + Z_1 + h_1)(Z_2^* + Z_2 + h_2)$. Write $a = Z_1 + h_1$, $b = Z_2 + h_2$, and let $s = \sqrt{1-\rho^2}$. A direct calculation gives

$$F_h(0 \mid Z) = \begin{cases} \Phi(Z_1 + (h_1 \vee h_2)) - \Phi(Z_1 + (h_1 \wedge h_2)) & \rho = 1 \\ \Phi(-a) + \Phi(-b) - 2\Phi_2(-a, -b; \rho) & \rho \in (-1, 1) \\ 1 + \Phi(Z_1 + (h_1 \wedge (-h_2))) - \Phi(Z_1 + (h_1 \vee (-h_2))) & \rho = -1 \end{cases}, \tag{C.20}$$

where $\Phi_2(\cdot, \cdot; \rho)$ is the standard bivariate normal CDF with correlation ρ.

Now consider the case $\rho = \pm 1$. Fix $x > y$, and for $z \in \mathbb{R}$, define

$$G_{x,y}(z) = \Phi(z + x) - \Phi(z + y). \tag{C.21}$$

Differentiation gives

$$\frac{\partial G_{x,y}(z)}{\partial z} = \phi(z + x) - \phi(z + y). \tag{C.22}$$

It is zero exactly when $z + x = -z - y$, strictly negative when $z + x > -z - y$, and strictly positive when $z + x < -z - y$. Thus, the level set $\{z : G_{x,y}(z) = c\}$ contains at most two points and hence has Lebesgue measure zero. If $|h_1| \neq |h_2|$, we have $h_1 \vee h_2 > h_1 \wedge h_2$ and $h_1 \vee (-h_2) > h_1 \wedge (-h_2)$. Then the preceding level-set argument implies that the distributions of $G_{h_1 \vee h_2, h_1 \wedge h_2}(Z_1)$ and $G_{h_1 \vee (-h_2), h_1 \wedge (-h_2)}(Z_1)$ have no atoms. This shows that, when $\rho = \pm 1$ and $|h_1| \neq |h_2|$, $F_h(0|Z)$ has no atoms.

For $\rho \in (-1, 1)$, let

$$G(a, b) := \Phi(-a) + \Phi(-b) - 2\Phi_2(-a, -b; \rho). \tag{C.23}$$

For fixed b, differentiation gives

$$\frac{\partial G(a, b)}{\partial a} = \phi(a) \left[2\Phi\left(\frac{\rho a - b}{s} \right) - 1 \right]. \tag{C.24}$$

We now make the level-set argument explicit. Suppose first that $\rho \neq 0$. The derivative is zero exactly when

$$\rho a - b = 0, \qquad \text{that is,} \qquad a = \frac{b}{\rho}. \tag{C.25}$$

If $\rho > 0$ ($\rho < 0$), it is strictly negative (positive) when $a < b/\rho$ and strictly positive (negative) when $a > b/\rho$. Thus $a \mapsto G(a, b)$ is strictly decreasing (increasing) on $(-\infty, b/\rho)$ and strictly increasing (decreasing) on

$(b/\rho, \infty)$. Therefore, for every c, the level set

$$\{a : G(a, b) = c\} \tag{C.26}$$

contains at most two points, and hence has Lebesgue measure zero.

If $\rho = 0$, then

$$\frac{\partial G(a,b)}{\partial a} = \phi(a)\big[2\Phi(-b) - 1\big]. \tag{C.27}$$

For $b \neq 0$, this derivative has a constant nonzero sign, so $G(\cdot, b)$ is strictly monotone and every level set contains at most one point and hence has Lebesgue measure zero. When $b = 0$, $G(a, 0) = 1/2$ for all a; but the event $b = 0$ has probability zero because $b = Z_2 + h_2 \sim N(h_2, 1)$ has a continuous distribution. Conditional on every nonexceptional value of b, $a = Z_1 + h_1$ has a continuous conditional distribution. Fubini's theorem therefore implies

$$\Pr(F_h(0 \mid Z) = c) = 0 \quad \text{for every } c \in \mathbb{R}. \tag{C.28}$$

That is, $F_h(0|Z)$ has no atom. □

Lemma 6. *Suppose that Assumption 1 holds. Let c_n be a sequence in $\mathbb{R}$ such that $c_n \to \infty$ as $n \to \infty$. Then $\Pr(Z^*_{jn} + Z_{jn} \leq c_n|\mathcal{F}_n) \to_p 1$ and $\Pr(Z^*_{jn} + Z_{jn} \leq -c_n|\mathcal{F}_n) \to_p 0$, for $j = 1, 2$.*

Proof of Lemma 6. By the same arguments as those for Lemma 3, the conditional law of $(Z^*_{jn} + Z_{jn})$ given $\mathcal{F}_n$ converges in distribution to the conditional law of $(Z^*_j + Z_j)$ given Z_j, for $j = 1, 2$. And by the same arguments as those for (C.17) in the proof of Lemma 4 and those for Lemma 5, we can show that, for all $c \in \mathbb{R}$,

$$\begin{aligned} &\Pr(Z^*_{jn} + Z_{jn} \leq c|\mathcal{F}_n) \to_d \Pr(Z^*_j + Z_j \leq c|Z_j) \equiv \Phi(c - Z_j) \text{ and} \\ &\Pr(Z^*_{jn} + Z_{jn} < c|\mathcal{F}_n) \to_d \Pr(Z^*_j + Z_j \leq c|Z_j) = \Phi(c - Z_j). \end{aligned} \tag{C.29}$$

Consider an arbitrary small $\varepsilon > 0$. Let c_ε be such that

$$\Pr(\Phi(c_\varepsilon - Z_j) < 1 - \varepsilon) < \varepsilon. \tag{C.30}$$

Such a c_ε always exists because $Z_j \sim N(0, 1)$ and hence $\Phi(c - Z_j) \to 1$ almost surely as $c \to \infty$. Since

$c_n \to \infty$, we have that $c_n > c_\varepsilon$ eventually. Thus, eventually

$$\Pr(Z^*_{jn} + Z_{jn} \leq c_n | \mathcal{F}_n) \geq \Pr(Z^*_{jn} + Z_{jn} \leq c_\varepsilon | \mathcal{F}_n) \to_d \Phi(c_\varepsilon - Z_j), \tag{C.31}$$

where the convergence holds by the first line of (C.29).

Now consider an arbitrary $\eta > 0$, and let ε be an arbitrary value such that $\varepsilon \in (0, (\eta \wedge 1))$. We have

$$\begin{aligned}
\Pr\left(|1 - \Pr(Z^*_{jn} + Z_{jn} \leq c_n | \mathcal{F}_n)| > \eta\right) &= \Pr\left(\Pr(Z^*_{jn} + Z_{jn} \leq c_n | \mathcal{F}_n) < 1 - \eta\right) \\
&\leq \Pr\left(\Pr(Z^*_{jn} + Z_{jn} \leq c_n | \mathcal{F}_n) < 1 - \varepsilon\right) \\
&\leq \Pr\left(\Pr(Z^*_{jn} + Z_{jn} \leq c_\varepsilon | \mathcal{F}_n) < 1 - \varepsilon\right) \\
&\to \Pr(\Phi(c_\varepsilon - Z_j) < 1 - \varepsilon) \\
&< \varepsilon,
\end{aligned} \tag{C.32}$$

where the convergence holds because $\Phi(c_\varepsilon - Z_j)$ has a continuous distribution and hence $1-\varepsilon$ is a continuity point of its CDF. This implies that

$$\limsup_{n\to\infty} \Pr\left(|1 - \Pr(Z^*_{jn} + Z_{jn} \leq c_n | \mathcal{F}_n)| > \eta\right) < \varepsilon. \tag{C.33}$$

Taking limit as $\varepsilon \to 0$ on both sides gives us

$$\limsup_{n\to\infty} \Pr\left(|1 - \Pr(Z^*_{jn} + Z_{jn} \leq c_n | \mathcal{F}_n)| > \eta\right) = 0. \tag{C.34}$$

This proves the first statement of the lemma. The second statement follows from the same arguments after observing that $-Z^*_{jn}$ has the same limit conditional law as Z^*_{jn}, and $-Z_j$ has the same distribution as Z_j, and swapping the first line in (C.29) for the second line. □

Define the extension of the bivariate normal CDF $\Phi_2(a, b, \rho) : \mathbb{R}^2 \times (-1, 1) \mapsto (0, 1)$ to $\mathbb{R}^2 \times [-1, 1]$ by

$$\overline{\Phi}_2(a, b; \rho) := \begin{cases} \left[\Phi(a) - \Phi(-b)\right]_+, & \rho = -1, \\ \Phi_2(a, b; \rho), & -1 < \rho < 1, \\ \Phi(a \wedge b), & \rho = 1, \end{cases} \tag{C.35}$$

where $[x]_+ := \max\{x, 0\}$.

Lemma 7. *The map $(a, b, \rho) \mapsto \overline{\Phi}_2(a, b; \rho)$ is continuous on $\mathbb{R}^2 \times [-1, 1]$.*

Proof of Lemma 7. Consider a sequence $(a_n, b_n, \rho_n) \to (a, b, \rho)$ in $\mathbb{R}^2 \times [-1, 1]$ as $n \to \infty$. Let X, E be independent $N(0, 1)$ random variables, and define

$$Y_n := \rho_n X + \sqrt{1 - \rho_n^2}\, E, \qquad Y := \rho X + \sqrt{1 - \rho^2}\, E. \tag{C.36}$$

Then $Y_n \to Y$ almost surely. Moreover, (X, Y_n) is standard bivariate normal with correlation ρ_n, allowing for $Y_n = \pm X$ when $\rho_n \in \{-1, 1\}$. Thus,

$$\overline{\Phi}_2(a_n, b_n; \rho_n) = \Pr(X \le a_n,\ Y_n \le b_n). \tag{C.37}$$

Outside the null event $\{X = a\} \cup \{Y = b\}$, the indicators $1\{X \le a_n, Y_n \le b_n\}$ converge almost surely to $\mathbf{1}\{X \le a,\, Y \le b\}$. Dominated convergence therefore gives

$$\overline{\Phi}_2(a_n, b_n; \rho_n) \to \Pr(X \le a,\, Y \le b) = \overline{\Phi}_2(a, b; \rho), \tag{C.38}$$

which proves the claim. □

Lemma 8. *Let $(Z_{1,\rho}, Z_{2,\rho})$ be standard bivariate normal with correlation ρ. Fix $h_2 \in \mathbb{R}$. As $\rho \to 1^-$,*

$$\Phi(Z_{1,\rho}) + \Phi(Z_{2,\rho} + h_2) - 2\Phi_2(Z_{1,\rho}, Z_{2,\rho} + h_2; \rho) \ \Rightarrow\ \left|\Phi(X) - \Phi(X + h_2)\right|,$$

where $X \sim N(0, 1)$.

Proof of Lemma 8. Let X, E be independent $N(0, 1)$ random variables and work with the version

$$Z_{1,\rho} = X, \qquad Z_{2,\rho} = \rho X + \sqrt{1 - \rho^2}\, E. \tag{C.39}$$

As $\rho \to 1^-$, we have $(Z_{1,\rho}, Z_{2,\rho}, \rho) \to_d (X, X, 1)$.

By Lemma 7, the function

$$g(a, b, r) := \Phi(a) + \Phi(b + h_2) - 2\overline{\Phi}_2(a, b + h_2; r) \tag{C.40}$$

is continuous. Therefore, the continuous mapping theorem gives

$$g(Z_{1,\rho}, Z_{2,\rho}, \rho) \to_d g(X, X, 1) = \left|\Phi(X) - \Phi(X + h_2)\right|. \tag{C.41}$$

□

Lemma 9. *Let $(Z_{1,\rho}, Z_{2,\rho})$ be standard bivariate normal with correlation ρ. Fix $h_2 \in \mathbb{R}$. As $\rho \to -1^+$,*

$$\Phi(Z_{1,\rho}) + \Phi(Z_{2,\rho} + h_2) - 2\Phi_2(Z_{1,\rho}, Z_{2,\rho} + h_2; \rho) \;\Rightarrow\; 1 - |\Phi(X) - \Phi(X - h_2)|,$$

where $X \sim N(0,1)$.

Proof of Lemma 9. Let X, E be independent $N(0,1)$ random variables and work with the version

$$Z_{1,\rho} = X, \qquad Z_{2,\rho} = \rho X + \sqrt{1-\rho^2}\, E. \tag{C.42}$$

As $\rho \to -1^+$, $(Z_{1,\rho}, Z_{2,\rho}, \rho) \to_d (X, -X, -1)$.

Define $g(a,b,r) := \Phi(a) + \Phi(b + h_2) - 2\overline{\Phi}_2(a, b + h_2; r)$. This function is continuous by Lemma 7. Hence, the continuous mapping theorem gives

$$\begin{aligned} g(Z_{1,\rho}, Z_{2,\rho}, \rho) \to_d g(X, -X, -1) &= \Phi(X) + \Phi(-X + h_2) - 2[\Phi(X) - \Phi(X - h_2)]_+ \\ &= 1 + \Phi(X) - \Phi(X - h_2) - 2[\Phi(X) - \Phi(X - h_2)]_+ \\ &= 1 - |\Phi(X) - \Phi(X - h_2)|, \end{aligned} \tag{C.43}$$

where the second equality holds because $\Phi(-x) = 1 - \Phi(x)$ and the third equality uses $x - 2[x]_+ = -|x|$. □

Lemma 10 (quantile regularity)**.** *Let Y be a random vector in $\mathbb{R}^d$ with an everywhere positive density where d is a positive integer. Let $g : \mathbb{R}^d \to \mathbb{R}$ be a continuous function and let $V = g(Y)$. Let $F_V(x) = \Pr(V \le x)$ for $x \in \mathbb{R}$. Let $Q_V(y) = \inf\{x \in \mathbb{R} : F_V(x) \ge y\}$ for $y \in (0,1)$. Consider a $p \in (0,1)$.*

(a) *$F_V(c) < p$ for $c < Q_V(p)$ and $F_V(c) > p$ for $c > Q_V(p)$;*

(b) *Consider a sequence of real random variables $\{V_n : n \ge 1\}$ such that $V_n \to_d V$. Also consider a sequence $p_n \in (0,1) : n \ge 1$ such that $p_n \to p$. Then $Q_{V_n}(p_n) \to Q_V(p)$.*

Proof of Lemma 10. (a) $F_V(c) < p$ for $c < Q_V(p)$ by definition, and $F_V(Q_V(p)) \ge p$ by right continuity. Let $c > Q_V(p)$. If $F_V(c) = 1$, then $F_V(c) > p$. Otherwise $\Pr(V > c) > 0$. The range $I = g(\mathbb{R}^d)$ is an interval because g is continuous. This interval must contain a point in $(-\infty, Q_V(p)]$ because $F_V(Q_V(p)) \ge p > 0$. It must also contain a point in (c, ∞), because $\Pr(V > c) > 0$. Hence $(Q_V(p), c) \subset I$. The preimage of $(Q_V(p), c)$ is nonempty because $(Q_V(p), c) \subset I$ and open because g is continuous. Thus, it has positive probability since Y has everywhere positive density on $\mathbb{R}^d$. Thus, $F_V(c) \ge F_V(Q_V(p)) + \Pr(Q_V(p) < V < c) > p$.

(b) Fix $\eta > 0$. The CDF F_V has at most countably many discontinuities because it is weakly increasing. Then we can choose continuity points $c_- \in (Q_V(p) - \eta, Q_V(p))$ and $c_+ \in (Q_V(p), Q_V(p) + \eta)$. By (a), $F_V(c_-) < p < F_V(c_+)$. Since $F_{V_n}(c_\pm) \to F_V(c_\pm)$ and $p_n \to p$, for all large enough n we have $F_{V_n}(c_-) < p_n < F_{V_n}(c_+)$. The second inequality gives $Q_{V_n}(p_n) \le c_+$. The first implies that $Q_{V_n}(p_n) \ge c_-$. Thus $|Q_{V_n}(p_n) - Q_V(p)| < \eta$ for all large enough n. Since η was arbitrary, this proves $Q_{V_n}(p_n) \to Q_V(p)$. □

Lemma 11. *The CDF $F_\rho(c; x, y)$ defined in (76) is continuous in (ρ, c, x, y) on $[-1, 1] \times \mathbb{R}^3$, and $w_\rho(x, y)$ defined in (77) is jointly continuous in (ρ, x, y) on $[-1, 1] \times \mathbb{R}^2$.*

Proof. We first show that for each $(\rho, x, y) \in [-1, 1] \times \mathbb{R}^2$, $F_\rho(\cdot; x, y)$ is continuous. Let N and M be independent standard normal random variables. Let $Y = (N, M)'$. Then $V_{\rho,x,y} := (N+x)(\rho N + \sqrt{1-\rho^2}M + y)$ is continuous in (ρ, x, y, Y) and $F_\rho(c; x, y) = \Pr(V_{\rho,x,y} \le c)$. For fixed (ρ, x, y) and any $c \in \mathbb{R}$, $V_{\rho,x,y} - c$ is a nonconstant polynomial in Y, because $\rho^2 + (1 - \rho^2) = 1$ forces its N^2 or NM coefficient to be nonzero. Thus, its zero set has Lebesgue measure zero. This implies that $F_\rho(\cdot; x, y)$ is continuous.

Along a convergent sequence $(\rho_n, x_n, y_n, c_n) \to (\rho, x, y, c) \in [-1, 1] \times \mathbb{R}^3$, $V_{\rho_n,x_n,y_n} - c_n \to V_{\rho,x,y} - c$ almost surely. This implies that $V_{\rho_n,x_n,y_n} - c_n \to_d V_{\rho,x,y} - c$. The continuity of $F_\rho(\cdot; x, y)$ implies that $\Pr(V_{\rho,x,y} - c = 0) = 0$. Thus, the convergence of $V_{\rho_n,x_n,y_n} - c_n$ implies that $\Pr(V_{\rho_n,x_n,y_n} - c_n \le 0) \to \Pr(V_{\rho,x,y} - c \le 0)$. That is, $F_{\rho_n}(c_n; x_n, y_n) \to F_\rho(c; x, y)$, proving the continuity of $F_\rho(c; x, y)$ in (ρ, c, x, y).

Again consider a convergent sequence $(\rho_n, x_n, y_n) \to (\rho, x, y) \in [-1, 1] \times \mathbb{R}^2$. Then $V_{\rho_n,x_n,y_n} \to V_{\rho,x,y}$ almost surely, which implies that $V_{\rho_n,x_n,y_n} \to_d V_{\rho,x,y}$. By the continuity of b (defined in (77)), we have $b_n := b(x_n, y_n) \to b(x, y) =: b$ and $1 - b_n \to 1 - b$. Equation (78) implies that $b \in (0, 1)$. Then, Lemma 10(b) gives the convergence of both quantiles; hence, of the maximum defining $w_{\rho_n}(x_n, y_n)$. □

Lemma 12 (symmetries)**.** *For any $x, y \in \mathbb{R}$, $w_\rho(-x, -y) = w_\rho(x, y)$ and $w_\rho(x, y) = w_{-\rho}(x, -y)$. Hence $a_\alpha(\rho, -h_2) = a_\alpha(\rho, h_2) = a_\alpha(-\rho, h_2)$ for all $\rho \in [-1, 1]$, $h_2 \in \mathbb{R}$.*

Proof. Fix an arbitrary $\rho \in [-1, 1]$ and $x, y, h_2 \in \mathbb{R}$.

Since $-Z^* \overset{d}{=} Z^*$, $F_\rho(c; -x, -y) = F_\rho(c; x, y)$. Also $b(-x, -y) = b(x, y) = b(x, -y)$. This proves the first identity: $w_\rho(-x, -y) = w_\rho(x, y)$.

For the second, note that $\widetilde{Z}^* = (Z_1^*, -Z_2^*)$ has covariance $\Sigma_{-\rho} = \left(\begin{smallmatrix} 1 & -\rho \\ -\rho & 1 \end{smallmatrix}\right)$, and $(Z_1^* + x)(Z_2^* + y) = -(\tilde{Z}_1^* + x)(\tilde{Z}_2^* - y)$. Consider the derivation

$$\begin{aligned} F_\rho(c; x, y) &= \Pr((Z_1^* + x)(Z_2^* + y) \le c) = \Pr(-(\tilde{Z}_1^* + x)(\tilde{Z}_2^* - y) \le c) \\ &= 1 - \Pr((\tilde{Z}_1^* + x)(\tilde{Z}_2^* - y) < -c) \\ &= 1 - F_{-\rho}(-c; x, -y), \end{aligned} \tag{C.44}$$

where the last equality holds because $F_{-\rho}(\cdot; x, -y)$ is continuous by Lemma 11.

To simplify notation, let $G(\cdot) = F_{-\rho}(\cdot; x, -y)$. Fix a $p \in (0,1)$. Let $\xi = Q_{-\rho}(1-p; x, -y)$ be the $(1-p)$-quantile of G. By Lemma 10(a) and continuity of G, $G(s) < 1-p$ for $s < \xi$, $G(\xi) = 1-p$, and $G(s) > 1-p$ for $s > \xi$. Hence

$$F_\rho(c; x, y) \geq p \iff G(-c) \leq 1-p \iff -c \leq \xi \iff c \geq -\xi. \tag{C.45}$$

This implies that $Q_\rho(p; x, y) = -\xi = -Q_{-\rho}(1-p; x, -y)$. Applying this with $p = b := b(x, y)$ and $p = 1-b$, and using $b(x, -y) = b(x, y)$, gives

$$\begin{aligned} w_{-\rho}(x, -y) &= \max\{Q_{-\rho}(b(x,-y); x, -y), -Q_{-\rho}(1 - b(x,-y); x, -y)\} \\ &= \max\{-Q_\rho(1 - b(x,y); x, y), Q_\rho(b(x,y); x, y)\} = w_\rho(x, y). \end{aligned} \tag{C.46}$$

Thus, the second identity in the lemma holds.

It remains to transfer these identities to a_α. Since $-Z \overset{d}{=} Z$, the first identity for w gives

$$W_{\rho,-h_2} = w_\rho(Z_1, Z_2 - h_2) = w_\rho(-Z_1, -Z_2 + h_2) \overset{d}{=} w_\rho(Z_1, Z_2 + h_2) = W_{\rho,h_2}. \tag{C.47}$$

Let $\tilde{Z} = (Z_1, -Z_2) \sim N(\mathbf{0}, \Sigma_{-\rho})$. The second identity gives

$$W_{\rho,h_2} = w_\rho(Z_1, Z_2 + h_2) = w_{-\rho}(Z_1, -Z_2 - h_2) = w_{-\rho}(\tilde{Z}_1, \tilde{Z}_2 - h_2) \overset{d}{=} W_{-\rho,-h_2}. \tag{C.48}$$

Variables with the same law have the same q-quantile. Thus, (C.47) and (C.48) imply that $a_\alpha(\rho, -h_2) = a_\alpha(\rho, h_2)$ and $a_\alpha(\rho, h_2) = a_\alpha(-\rho, -h_2)$, respectively. Applying the first equality at $-\rho$ gives $a_\alpha(-\rho, -h_2) = a_\alpha(-\rho, h_2)$, hence $a_\alpha(\rho, h_2) = a_\alpha(-\rho, h_2)$. □

Lemma 13. *The function a_α defined in (79) is jointly continuous on $[-1,1] \times \mathbb{R}$.*

Proof. Let N and M be independent standard normal random variables and $Z = (N, \rho N + \sqrt{1-\rho^2}M)$, so that $W_{\rho,h_2} = w_\rho(N, \rho N + \sqrt{1-\rho^2}M + h_2)$. By Lemma 11, this is continuous in (ρ, h_2, N, M). Hence $W_{\rho_n,h_{2n}} \to W_{\rho,h_2}$ almost surely along any converging sequence $(\rho_n, h_{2n}) \to (\rho, h_2)$ in $[-1,1] \times \mathbb{R}$, and W_{ρ,h_2} is a continuous function of (N, M). Lemma 10(b) with $p_n = q$ gives the convergence of the q-quantiles of $W_{\rho_n,h_{2n}}$ to that of W_{ρ,h_2}. The result follows since the positive part function is continuous. □

Lemma 14 (Uniform Nuisance-tail Control)**.** *For every $\varepsilon > 0$ there is $H_\varepsilon < \infty$, depending only on α, ε, such that $\Pr(W_{\rho,h_2} \leq \varepsilon) \geq q$, and hence $a_\alpha(\rho, h_2) \leq \varepsilon$, for all $\rho \in [-1,1]$ and $|h_2| \geq H_\varepsilon$.*

Proof of Lemma 14. Let $k = \Phi^{-1}(1-\alpha/2)$, $m = \phi(k+1)/4$, and $\kappa = \min\{1/4,\ 2m/\phi(0)\}$. Since $h_2\,\Phi(-h_2/2) \to 0$, we can choose a large enough H_ε depending only on α, ε such that $h_2\Phi(-h_2/2) < \kappa m\varepsilon$ for all $h_2 \geq H_\varepsilon$. We choose a H_ε that also satisfies $H_\varepsilon \geq \max\{\varepsilon, 2(k+2)\}$.

By Lemma 12, $W_{\rho,-h_2}$ and W_{ρ,h_2} have the same law. Thus, it suffices to take $h_2 \geq H_\varepsilon$. Let $\delta = m\varepsilon/(2\phi(0)h_2)$; then $\delta \leq 1/8$, since $m/(2\phi(0)) \leq 1/8$ and $\varepsilon \leq h_2$. Define

$$E = \{|Z_1| \leq k+\delta,\ |Z_2| \leq h_2/2\}. \tag{C.49}$$

Fix $(z_1, z_2) = (Z_1(\omega), Z_2(\omega))$ for $\omega \in E$ and let $y = z_2 + h_2$. Then $y \in [h_2/2, 3h_2/2]$, which implies that $y > 0$ and $|1 - 2\Phi(y)| = 1 - 2\Phi(-y) \geq 1 - 2\Phi(-h_2/2)$. The maximum in the definition of $b(\cdot,\cdot)$ (in (77)) is at least this last term. Thus

$$b(z_1, y) \leq \tfrac{1}{2}\big(1 - q(1 - 2\Phi(-h_2/2))\big) = \alpha/2 + q\Phi(-h_2/2). \tag{C.50}$$

Let $V = (Z_1^* + z_1)(Z_2^* + y)$ with CDF F and quantile function Q. Then

$$\begin{aligned} F(0) &= \Pr(Z_1^* + z_1 \leq 0, Z_2^* + y > 0) + \Pr(V \leq 0, Z_2^* + y \leq 0) \\ &= \Pr(Z_1^* + z_1 \leq 0) - \Pr(Z_1^* + z_1 \leq 0, Z_2^* + y \leq 0) + \Pr(V \leq 0, Z_2^* + y \leq 0) \\ &= \Phi(-z_1) - \Pr(Z_1^* + z_1 \leq 0, Z_2^* + y \leq 0) + \Pr(V \leq 0, Z_2^* + y \leq 0). \end{aligned} \tag{C.51}$$

Thus,

$$|F(0) - \Phi(-z_1)| \leq \Pr(Z_2^* + y \leq 0) = \Phi(-y) \leq \Phi(-h_2/2). \tag{C.52}$$

Let $S_+ = \{0 < Z_1^* + z_1 \leq \varepsilon/(2h_2),\ |Z_2^*| \leq h_2/2\}$ and $S_- = \{-\varepsilon/(2h_2) < Z_1^* + z_1 < 0,\ |Z_2^*| \leq h_2/2\}$. On both, $0 \leq Z_2^* + y \leq 2h_2$. Then, $S_+ \subseteq \{0 \leq V \leq \varepsilon\}$ and $S_- \subseteq \{-\varepsilon < V \leq 0\}$. On both strips,

$$|Z_1^*| \leq |z_1| + \varepsilon/(2h_2) \leq k + \delta + \varepsilon/(2h_2) \leq k+1. \tag{C.53}$$

Given $Z_1^* = x$ with $|x| \leq k+1$, $Z_2^* \sim N(\rho x, 1-\rho^2)$. Since $|\rho| \leq 1$ and $h_2 \geq 2(k+2)$, we have $|\rho x| \leq k+1 \leq h_2/2 - 1$. Thus, $|Z_2^* - \rho x| \leq 1 \Rightarrow |Z_2^*| \leq h_2/2$. This implies

$$\begin{aligned} \Pr(|Z_2^*| \leq h_2/2 \mid Z_1^* = x) &\geq \Pr(|Z_2^* - \rho x| \leq 1 \mid Z_1^* = x) \\ &= \Pr(|N(0, 1-\rho^2)| \leq 1) \geq \Pr(|N(0,1)| \leq 1) > \tfrac{1}{2} \end{aligned} \tag{C.54}$$

uniformly in ρ, including $\rho = \pm 1$. Now S_+ requires $Z_1^* \in (-z_1, -z_1 + \varepsilon/(2h_2)]$, an interval of length $\varepsilon/(2h_2)$ on which $|Z_1^*| \le k+1$. Integrating the conditional bound over this interval,

$$\Pr(S_+) = \int_{-z_1}^{-z_1+\varepsilon/(2h_2)} \phi(x) \Pr(|Z_2^*| \le h_2/2 \mid Z_1^* = x)\, dx \ge \phi(k+1) \cdot \frac{\varepsilon}{2h_2} \cdot \frac{1}{2} = \frac{m\varepsilon}{h_2}, \tag{C.55}$$

where $\phi(x) \ge \phi(k+1)$ is implied by (C.53). The same argument applies to S_-, whose Z_1^*-interval is $(-z_1 - \varepsilon/(2h_2), -z_1)$, giving

$$\Pr(S_-) \ge \frac{m\varepsilon}{h_2}. \tag{C.56}$$

On E, we have $|z_1| \le k + \delta$ and $\phi(x) \le \phi(0)$ for any $x \in \mathbb{R}$. Also recall that $\Phi(k) = 1 - \alpha/2$. Thus,

$$\alpha/2 - \phi(0)\delta \le \Phi(-k-\delta) \le \Phi(-z_1) \le \Phi(k+\delta) \le 1 - \alpha/2 + \phi(0)\delta. \tag{C.57}$$

Now since $\phi(0)\delta = m\varepsilon/(2h_2)$, combining this with (C.52), (C.55) and (C.56), we have

$$\begin{aligned} F(\varepsilon) &= F(0) + \Pr(0 \le V \le \varepsilon) \ge F(0) + \Pr(S_+) \ge \alpha/2 + \tfrac{m\varepsilon}{2h_2} - \Phi(-h_2/2), \\ F(-\varepsilon) &= F(0) - \Pr(-\varepsilon < V \le 0) \le F(0) - \Pr(S_-) \le 1 - \alpha/2 - \tfrac{m\varepsilon}{2h_2} + \Phi(-h_2/2), \end{aligned} \tag{C.58}$$

where the first equality in the first line holds by the continuity of $F(\cdot)$ established in Lemma 11. Since $h_2\Phi(-h_2/2) < m\varepsilon/4 \le m\varepsilon/(2(1+q))$, we have $\frac{m\varepsilon}{2h_2} - \Phi(-h_2/2) > q\Phi(-h_2/2)$. Thus,

$$\begin{aligned} F(\varepsilon) &> \alpha/2 + q\Phi(-h_2/2) \ge b(z_1, y) \\ F(-\varepsilon) &< 1 - \alpha/2 - q\Phi(-h_2/2) \le 1 - b(z_1, y), \end{aligned} \tag{C.59}$$

where the second inequality in each line follows from (C.50). The first gives $Q(b(z_1, y)) \le \varepsilon$. The second gives $F(c) < 1 - b(z_1, y)$ for all $c \le -\varepsilon$, so $Q(1 - b(z_1, y)) \ge -\varepsilon$. Thus, $w_\rho(z_1, z_2 + h_2) \le \varepsilon$. Therefore $E \subset \{W_{\rho,h_2} \le \varepsilon\}$.

Now we calculate the probability of E, where (Z_1, Z_2) is the random vector in the definition of W_{ρ,h_2}. Each of Z_1, Z_2 is marginally standard normal for every $\rho \in [-1, 1]$. The complement of E is $\{|Z_1| > k + \delta\} \cup \{|Z_2| > h_2/2\}$. Thus by the union bound for probability measures, we have

$$\Pr(E) \ge 1 - \Pr(|Z_1| > k + \delta) - \Pr(|Z_2| > h_2/2) = 1 - 2\Phi(-k-\delta) - 2\Phi(-h_2/2). \tag{C.60}$$

Since $\Phi(-k) = \alpha/2$, we have $1 - 2\Phi(-k) = q$. Thus,

$$1 - 2\Phi(-k-\delta) = q + 2\big(\Phi(-k) - \Phi(-k-\delta)\big) = q + 2\int_k^{k+\delta} \phi(x)\,dx. \tag{C.61}$$

Since $\delta < 1$, $\phi(x) \geq \phi(k+1)$ for $x \in [k, k+\delta]$ (as ϕ is decreasing on $[0,\infty)$), the integral is at least $\delta\phi(k+1)$. Substituting $\delta = m\varepsilon/(2\phi(0)h_2)$,

$$\Pr(E) \geq q + 2\delta\phi(k+1) - 2\Phi(-h_2/2) = q + \frac{\phi(k+1)\,m\varepsilon}{\phi(0)\,h_2} - 2\Phi(-h_2/2) \geq q. \tag{C.62}$$

The last inequality in (C.62) holds because

$$h_2\Phi(-h_2/2) < \kappa m\varepsilon \leq \frac{2m^2\varepsilon}{\phi(0)} = \frac{\phi(k+1)\,m\varepsilon}{2\phi(0)}, \text{ for } h_2 \geq H_\varepsilon, \tag{C.63}$$

where the first inequality holds by the definition of H_ε, the second inequality holds by $\kappa \leq 2m/\phi(0)$, and the equality holds by $\phi(k+1) = 4m$.

Since $E \subset \{W_{\rho,h_2} \leq \varepsilon\}$, we get $\Pr(W_{\rho,h_2} \leq \varepsilon) \geq \Pr(E) \geq q$ for every $\rho \in [-1,1]$ and $h_2 \geq H_\varepsilon$. Hence $Q_{W_{\rho,h_2}}(q) \leq \varepsilon$, and since $\varepsilon > 0$, $a_\alpha(\rho, h_2) = \max\{0, Q_{W_{\rho,h_2}}(q)\} \leq \varepsilon$. The bound is uniform in ρ because H_ε depends only on α and ε. □

Lemma 15. *Suppose $\rho \in [-1,1]$ and $h \in \mathbb{R}^2$. Then, for every $c \in \mathbb{R}$,*

$$\Pr\{F_h(c|Z) = \beta(\alpha)/2\} = \Pr\{F_h(c|Z) = 1 - \beta(\alpha)/2\} = 0. \tag{C.64}$$

Proof of Lemma 15. Let $q = 1-\alpha$ and $k = \Phi^{-1}(1-\alpha/2)$. First suppose $|\rho| < 1$. Recall that $F_h(c|Z) = F_\rho(c; Z_1+h_1, Z_2+h_2)$ and $\beta(\alpha)/2 = b(Z_1+h_1, Z_2+h_2)$. Since (Z_1+h_1, Z_2+h_2) has a density, it suffices to show that the sets $\{(x,y) : F_\rho(c;x,y) = b(x,y)\}$ and $\{(x,y) : F_\rho(c;x,y) = 1-b(x,y)\}$ have Lebesgue measure zero.

For fixed c, y, the function $x \mapsto F_\rho(c;x,y)$ is real analytic. To see this, let $\phi_2(x,y,\rho)$ denote the standard bivariate normal density with correlation ρ. Fix $x_0 \in \mathbb{R}$ and let $D = 1-\rho^2$ and $A(u,v) = (u - x_0 - \rho(v-y))/D$. For real t,

$$\phi_2(u - x_0 - t, v - y, \rho) = \phi_2(u - x_0, v - y, \rho)e^{tA(u,v) - t^2/(2D)}. \tag{C.65}$$

Expanding the exponential and integrating gives

$$\begin{aligned} F_\rho(c; x_0 + t, y) &= \int\int_{uv \le c} \phi_2(u - x_0 - t, v - y, \rho) du dv \\ &= e^{-t^2/(2D)} \int\int_{uv \le c} \phi_2(u - x_0, v - y, \rho) \sum_{m=0}^{\infty} \frac{t^m A(u,v)^m}{m!} du dv \\ &= e^{-t^2/(2D)} \sum_{m=0}^{\infty} \frac{t^m}{m!} \iint_{uv \le c} A(u,v)^m \phi_2(u - x_0, v - y, \rho)\, du\, dv, \end{aligned} \tag{C.66}$$

where the interchange of the integral and the infinite sum holds by dominated convergence. Dominated convergence holds because for any M,

$$\left| \sum_{m=0}^{M} \frac{t^m A(u,v)^m}{m!} \right| \le e^{|t||A(u,v)|}, \tag{C.67}$$

and under the density $\phi_2(u - x_0, v - y, \rho)$, $A(u,v)$ is normal with mean zero and variance $1/D$ which implies that $e^{|t||A(u,v)|}$ is integrable. This proves the real analyticity of $F_\rho(c; \cdot, y)$ at every x_0, for fixed ρ, c, y.

Moreover, dominated convergence directly applied on the representation $F_\rho(c; x, y) = \Pr((Z_1^* + x)(Z_2^* + y) \le c)$ for $(Z_1^*, Z_2^*)' \sim N\left(\mathbf{0}, \left(\begin{smallmatrix} 1 & \rho \\ \rho & 1 \end{smallmatrix}\right)\right)$ gives

$$\lim_{x \to \infty} F_\rho(c; x, y) = \Phi(-y), \qquad \lim_{x \to -\infty} F_\rho(c; x, y) = \Phi(y). \tag{C.68}$$

Fix $y \notin \{0, k, -k\}$. Let

$$M_y = \max\{q, |1 - 2\Phi(y)|\}, \qquad a_y = \Phi^{-1}((1 + M_y)/2). \tag{C.69}$$

Then

$$b(x,y) = \begin{cases} \alpha/2 + q\Phi(x), & x < -a_y, \\ (1 - qM_y)/2, & |x| \le a_y, \\ \alpha/2 + q\Phi(-x), & x > a_y. \end{cases} \tag{C.70}$$

Since Φ is real analytic, each formula defines a real-analytic function on all of $\mathbb{R}$.

Thus, the difference between $F_\rho(c; x, y)$ and the $\mathbb{R}$-extension of each piece of $b(x,y)$ is real analytic on $\mathbb{R}$. By the real-analytic identity theorem, agreement on any open interval would imply agreement everywhere. None of these differences is identically zero: by (C.68), an identity on an open subinterval of $(-a_y, a_y)$ would imply $\Phi(-y) = (1 - qM_y)/2 = \Phi(y)$, hence $y = 0$; an identity on an open interval in (a_y, ∞) would imply

$\Phi(-y) = \lim_{x\to\infty}(\alpha/2 + q\Phi(-x)) = \alpha/2$, hence $y = k$; and an identity on an open interval in $(-\infty, -a_y)$ would imply $\Phi(y) = \lim_{x\to-\infty}(\alpha/2 + q\Phi(x)) = \alpha/2$, hence $y = -k$. All three cases lead to a contradiction with $y \notin \{0, k, -k\}$.

The same argument applies to $F_\rho(c; x, y) - (1 - b(x, y))$: an identity on an open interval in the middle, right, or left interval would force $y = 0, -k, k$, respectively.

Thus, for each $y \notin \{0, k, -k\}$, the difference between $F_\rho(c; \cdot, y)$ and the real-analytic extension of each piece of $b(\cdot, y)$ or $1 - b(\cdot, y)$ is not identically zero. A nonzero real-analytic function on $\mathbb{R}$ has isolated zeros. Thus, for each $y \notin \{0, k, -k\}$, each equality holds for at most countably many values of x, possibly including the two breakpoints $-a_y$ and a_y.

Now we establish (C.64) by applying Tonelli's theorem. Let E denote either of the sets

$$\{(x, y) : F_\rho(c; x, y) = b(x, y)\}, \qquad \{(x, y) : F_\rho(c; x, y) = 1 - b(x, y)\}. \tag{C.71}$$

The set E is measurable because $F_\rho(c; x, y)$ and $b(x, y)$ are continuous in (x, y); the former follows by Lemma 11.

Write $E_y = \{x : (x, y) \in E\}$ for its horizontal section at y. For every $y \notin \{0, k, -k\}$, the section E_y is countable, so its one-dimensional Lebesgue measure is zero. The three exceptional values of y themselves form a Lebesgue-null set. Hence, writing λ_1 and λ_2 for one- and two-dimensional Lebesgue measure, Tonelli's theorem gives

$$\lambda_2(E) = \int_{\mathbb{R}} \left(\int_{\mathbb{R}} \mathbf{1}_E(x, y)\, dx \right) dy = \int_{\mathbb{R}} \lambda_1(E_y)\, dy = 0. \tag{C.72}$$

Although the countable set E_y may vary with y, its measure is zero for almost every y, which is all this integral requires. Since $(Z_1 + h_1, Z_2 + h_2)$ has a density, it assigns a probability of zero to E. Applying this argument to both equality sets proves (C.64) for $|\rho| < 1$.

Now suppose $\rho = 1$. Write $Z = (V, V)$ and $Z^* = (U, U)$, where U, V are independent standard normals. With $a = (h_1 + h_2)/2$ and $d = (h_1 - h_2)/2$,

$$T_h = (U + V + h_1)(U + V + h_2) = (U + V + a)^2 - d^2. \tag{C.73}$$

If $c + d^2 \le 0$, then $F_h(c|Z) = 0$, whereas both $\beta(\alpha)/2$ and $1 - \beta(\alpha)/2$ lie strictly between zero and one. The equalities in (C.64) are impossible, and hence (C.64) holds trivially.

If $c + d^2 > 0$, let $r = \sqrt{c + d^2}$. Then

$$F_h(c|Z) = \Phi(r - V - a) - \Phi(-r - V - a). \tag{C.74}$$

This is a real-analytic function of V and tends to zero as $V \to \pm\infty$. The function $b(V + h_1, V + h_2)$ can be written as

$$\begin{aligned}
&b(V + h_1, V + h_2) \\
&= (1 - q(q \vee |1 - 2\Phi(V + h_1)| \vee |1 - 2\Phi(V + h_2)|))/2 \\
&= \begin{cases} \alpha/2 + q\Phi(V + (h_1 \wedge h_2)) & \text{if } V < \Phi^{-1}(\alpha/2) - (h_1 \wedge h_2), V \le -(h_1 + h_2)/2 \\ \alpha/2 + q(1 - \Phi(V + (h_1 \vee h_2))) & \text{if } V > \Phi^{-1}(1 - \alpha/2) - (h_1 \vee h_2), V > -(h_1 + h_2)/2 \\ (1 - q^2)/2 & \text{otherwise.} \end{cases}
\end{aligned} \tag{C.75}$$

As we can see, $b(V + h_1, V + h_2)$ and $1 - b(V + h_1, V + h_2)$, as a function of V, have finitely many analytic pieces. Each piece has a real-analytic extension to $\mathbb{R}$.

If $F_h(c|Z) - b(V + h_1, V + h_2)$ is identically zero on an open interval, then $F_h(c|Z)$ must be identical to the $\mathbb{R}$-extension of one of the three pieces by the real-analyticity of these extensions and that of $F_h(c|Z)$. However, this is impossible because the limit of $F_h(c|Z)$ as $V \to \infty$ is zero, while the limit of the $\mathbb{R}$-extension of the three pieces are $1 - \alpha/2$, $\alpha/2$, and $(1 - q^2)/2$, respectively, none of which equal to zero.

Therefore, the difference between $F_h(c|Z)$ and the $\mathbb{R}$-extension of each piece of $b(V + h_1, V + h_2)$ is not identically zero. By the same argument, the difference between $F_h(c|Z)$ and the $\mathbb{R}$-extension of each piece of $1 - b(V + h_1, V + h_2)$ is not identically zero either.

Any nonzero real analytic function has only isolated zeros. Thus, the set of V values on which $F_h(c|Z) = b(V + h_1, V + h_2)$ or $F_h(c|Z) = 1 - b(V + h_1, V + h_2)$ is at most a countable set, possibly including the breakpoints $\Phi^{-1}(\alpha/2) - (h_1 \wedge h_2)$, $\Phi^{-1}(1 - \alpha/2) - (h_1 \vee h_2)$, and $-(h_1 + h_2)/2$. Thus, (C.64) follows from the fact that the distribution of V has no atom.

For $\rho = -1$, we can write $Z = (V, -V)$ and $Z^* = (U, -U)$, and

$$T_h = (U + V + h_1)(-U - V + h_2) = -(U + V + d)^2 + a^2, \tag{C.76}$$

where U and V are independent standard normal random variables and a and d are the same as in the $\rho = 1$ case. If $a^2 - c \le 0$, then $F_h(c|Z) = 1$, whereas both $\beta(\alpha)/2$ and $1 - \beta(\alpha)/2$ lie strictly between 0 and 1.

Thus, (C.64) holds trivially. If $a^2 - c > 0$, then

$$F_h(c|Z) = 1 - \Phi(\sqrt{a^2 - c} - V - d) + \Phi(-\sqrt{a^2 - c} - V - d), \tag{C.77}$$

which is a real-analytic function of V and tends to 1 as $V \to \pm\infty$. The function $b(V + h_1, -V + h_2) = b(V + h_1, V - h_2)$ can be written as a piecewise function in the same way as the $\rho = 1$ case except for the flipped sign of h_2. Then, the same arguments in the $\rho = 1$ case can show that (C.64) holds when $\rho = -1$ as well. □

D Numerical Procedure for Computing the Size Correction

We first simplify the numerical problem with a few theoretical observations.

For any $\rho \in [-1, 1]$ and h, define

$$f_{1-\alpha}(\rho, h) = \text{the } (1-\alpha) \text{ quantile of } \max\left\{F_h^{-1}\left(\tfrac{\beta(\alpha)}{2}|Z\right), -F_h^{-1}\left(1 - \tfrac{\beta(\alpha)}{2}|Z\right)\right\}. \tag{D.1}$$

The proof of Lemma 1 establishes that

$$f_{1-\alpha}(\rho) = \max\left\{0, \sup_{h:h_1h_2=0} f_{1-\alpha}(\rho; h)\right\}. \tag{D.2}$$

It is sufficient to consider $h = (0, h_2)$ where $h_2 \in \mathbb{R}$ because indices 1 and 2 are symmetric in our formulas. Also, notice that if we flip the signs of h_2 and ρ simultaneously, the new joint distribution of $F_h(x|Z)$ and $\beta(\alpha)$ coincides with the joint distribution of $1 - F_h(-x|Z)$ and $\beta(\alpha)$ before the sign switch, thanks to the symmetry of normal distributions. Thus, $f_{1-\alpha}(\rho) = f_{1-\alpha}(-\rho)$.

Using the formula in (D.1), we calculate the size correction term for each ρ by the following steps:

1. Fix ρ, α, a constant $0 < H < \infty$, and a grid $\mathcal{H} = \{h_{2,1}, \ldots, h_{2,K}\}$ such that $-H = h_{2,1} \leq \cdots \leq h_{2,K} = H$. Choose the number of Monte Carlo draws M and the number of conditional simulation draws B.

2. For each $h_2 \in \mathcal{H}$, generate M independent draws $(Z_{1,m}, Z_{2,m})$, $m = 1, \ldots, M$, from a bivariate normal distribution with mean zero, unit variances, and correlation coefficient ρ.

3. For each draw $(Z_{1,m}, Z_{2,m})$, compute

$$\beta_m(\alpha) = 1 - \Big[(1-\alpha) \vee \big\{|2\Phi(Z_{1,m}) - 1| \vee |2\Phi(Z_{2,m} + h_2) - 1|\big\}\Big](1-\alpha).$$

4. For each $(Z_{1,m}, Z_{2,m})$, generate B independent draws $(Z^*_{1,b}, Z^*_{2,b})$, $b = 1, \ldots, B$, from the same bivariate normal distribution, independently of $(Z_{1,m}, Z_{2,m})$. Then calculate

$$T^*_{m,b}(h_2) = (Z^*_{1,b} + Z_{1,m})(Z^*_{2,b} + Z_{2,m} + h_2).$$

5. Let $Q_m(\tau, h_2)$ denote the empirical τ quantile of $\{T^*_{m,b}(h_2) : b = 1, \ldots, B\}$. Compute

$$\mathcal{Q}_m(h_2) = \max\left\{Q_m\left(\frac{\beta_m(\alpha)}{2}, h_2\right), -Q_m\left(1 - \frac{\beta_m(\alpha)}{2}, h_2\right)\right\}.$$

6. Let $\hat{f}_{1-\alpha}(\rho; h_2)$ be the empirical $(1-\alpha)$ quantile of $\{\mathcal{Q}_m(h_2) : m = 1, \ldots, M\}$. Compute $f_{1-\alpha}(\rho)$ by maximizing over the grid:

$$\hat{f}_{1-\alpha}(\rho) = \max\left\{0, \max_{h_2 \in \mathcal{H}} \hat{f}_{1-\alpha}(\rho; h_2)\right\}.$$

We compute Table 2 using the above procedure with $M = 100,000$ and $B = 100,000$, and using the grid $\{0.01j : -100 \leq j \leq 100\}$ for h_2, corresponding to $H = 1$. We use $H = 1$ because preliminary searches find that larger H produce similar results. Even though $f_{1-\alpha}(\rho)$ and $f_{1-\alpha}(-\rho)$ are theoretically the same, we still calculate both. They can be numerically different because simulated normal distributions may not be exactly symmetric. The table reports the average of the calculated $f_{1-\alpha}(|\rho|)$ and $f_{1-\alpha}(-|\rho|)$.

E Correlation Formula for Simulation Example

Nonzero γ introduces dependence between the influence functions of $\widehat{\theta}_1$ and $\widehat{\theta}_2$, and hence leads to nonzero ρ. To see this, note that

$$\begin{aligned}
\widehat{\theta}_1 - \theta_1 &= \left(\frac{1}{n}\sum_{i=1}^n \dot{X}_i^2\right)^{-1} \frac{1}{n}\sum_{i=1}^n \dot{X}_i \dot{u}_i, \\
\widehat{\theta}_2 - \theta_2 &= \left(\frac{1}{n}\sum_{i=1}^n \ddot{M}_i^2\right)^{-1} \frac{1}{n}\sum_{i=1}^n \ddot{M}_i \ddot{e}_i,
\end{aligned} \tag{E.1}$$

where $\dot{X}_i = X_i - n^{-1}\sum_{i=1}^n X_i$, $\dot{u}_i = u_i - n^{-1}\sum_{i=1}^n u_i$, and $\ddot{M}_i$ and $\ddot{e}_i$ denote the residuals from projecting M_i and e_i on $(1, X_i)$, respectively. Since $M_i = \alpha_M + X_i\theta_1 + u_i$, the residualized regressor $\ddot{M}_i$ is asymptotically equivalent to u_i. Moreover, $\dot{X}_i$, $\dot{u}_i$, and $\ddot{e}_i$ are asymptotically equivalent to X_i, u_i, and e_i, respectively, because their expectations are zero and $E[X_i e_i] = 0$. Therefore, the asymptotic correlation between $\widehat{\theta}_1$ and

$\widehat{\theta}_2$ is

$$ACor(\widehat{\theta}_1, \widehat{\theta}_2) = \frac{E[X_i u_i^2 e_i]}{\sqrt{E[X_i^2 u_i^2] E[u_i^2 e_i^2]}}. \tag{E.2}$$

Note that $E[X_i^2 u_i^2] = E[X_i^2]E[u_i^2] = 1$ and

$$\begin{aligned} E[u_i^2 e_i^2] &= \frac{1}{2} E[u_i^2 e_i^2 || W_i| = 1] + \frac{1}{2} E[u_i^2 e_i^2 || W_i| = a] \\ &= \frac{\frac{1}{2} E[(a^2 + \tau^2 \varepsilon_i^2)(\gamma^2 X_i^2 + v_i^2)] + \frac{1}{2} E[(1 + \tau^2 \varepsilon_i^2)(\gamma^2 X_i^2 + v_i^2)]}{(1+\gamma^2)((1+a^2)/2 + \tau^2)} = 1. \end{aligned} \tag{E.3}$$

Therefore,

$$\begin{aligned} ACor(\widehat{\theta}_1, \widehat{\theta}_2) = E[X_i u_i^2 e_i] &= \frac{\gamma E[X_i^2 u_i^2 s(W_i)] + E[X_i u_i^2 v_i]}{\sqrt{1+\gamma^2}} = \frac{\gamma E[X_i^2 u_i^2 s(W_i)]}{\sqrt{1+\gamma^2}} \\ &= \frac{\gamma(\frac{1}{2} E[u_i^2 || W_i| = 1] - \frac{1}{2} E[u_i^2 || W_i| = a])}{\sqrt{1+\gamma^2}} = \frac{\gamma(\frac{1}{2} E[1 + \tau^2 \varepsilon_i^2] - \frac{1}{2} E[a^2 + \tau^2 \varepsilon_i^2])}{\sqrt{1+\gamma^2}((1+a^2)/2 + \tau^2)} \\ &= \frac{\gamma}{\sqrt{1+\gamma^2}} \frac{1-a^2}{1+a^2+2\tau^2}. \end{aligned} \tag{E.4}$$

Supplemental References

Billingsley, P. (1999). *Convergence of Probability Measures*. John Wiley & Sons, New York, 2 edition.